

Nearby Vibes

MVP Product Requirements Document

Version 1.8 · Status: Build-Ready — All Decisions Confirmed · Platform: Progressive Web App (PWA) · Prepared for: Engineering build-out via Claude · Document owner: Product Management · July 23, 2026

Contents

How to Read This Document	5
1. Executive Summary	7
2. Product Vision	8
3. Product Principles	9
4. MVP Definition	10
4.1 What the MVP validates	10
4.2 In scope for MVP	10
4.3 Explicitly out of scope for MVP	11
4.4 Target launch parameters	11
5. User Personas	12
5.1 Maya — The Nearby User	12
5.2 Sam — The Venue Owner/Manager	12
5.3 Jordan — The Delegated Venue Seat (Venue Team Member / Social Content Creator)	12
5.4 Alex — The Platform Admin (internal, Nearby Vibes staff)	13
6. User Stories	14
6.1 Maya (User) stories	14
6.2 Sam (Venue) stories	15
6.3 Jordan (Delegated Venue Seat) stories	16
6.4 Alex (Admin) stories	16
7. Functional Requirements	18
7.1 Accounts & Authentication (ACC)	18
7.2 Venue Onboarding & Claiming (ONB)	20
7.3 Discovery Experience — Tile & Map (UX)	21
7.4 Post / Special Lifecycle (POST)	23
7.5 Media Pipeline (MEDIA)	24
7.6 Navigation (NAV)	25
7.7 Static & Legal Pages (STATIC)	25
7.8 Admin (ADMIN)	25
7.9 SEO (SEO)	27
7.10 Push Notifications (PUSH)	27
7.11 QR (QR)	29
7.12 Subscription & Billing (SUB) — <i>New in v1.4</i>	29
8. Acceptance Criteria	32
8.1 Account creation	32

8.2 Venue claiming	32
8.3 Post lifecycle	32
8.4 Age gating	33
8.5 Discovery	33
8.6 Follow lifecycle	33
8.7 Push notifications	33
8.8 QR + arrival confirmation	33
8.9 Navigation tracking	34
8.10 Reporting	34
8.11 Dashboards	34
8.12 MVP “done” gate — original discovery/QR scope (from source concept brief, restated as acceptance criteria)	34
8.13 MVP “done” gate — Subscription & Billing (<i>new, v1.4</i>)	34
8.14 MVP “done” gate — Expanded Admin Panel (<i>new, v1.4</i>)	35
9. UI Specifications	36
9.1 Screen inventory	36
9.2 Home — Tile View	37
9.3 Home — Map View	38
9.4 Onboarding (first-run)	38
9.5 Age gate	38
9.6 Post creation wizard (Venue / Social Manager)	38
9.7 Venue dashboard	38
9.8 Subscription / Pricing panel (<i>new, v1.4</i>)	39
9.9 Checkout (<i>new, v1.4</i>)	39
9.10 QR landing	39
9.11 Admin portal login (<i>new, v1.4</i>)	39
9.12 Admin account management & onboarding queue (<i>new, v1.4</i>)	39
9.13 Footer (global)	40
9.14 Cookie consent banner (global)	40
9.15 Accessibility requirements (applies to all screens)	40
9.16 Visual design direction	40
10. Business Rules	42
10.1 Age gating (alcohol content)	42
10.2 Mandatory drink-special disclaimer (verbatim, immutable)	42
10.3 Post rules	42
10.4 Business email domain blocklist	43
10.5 Venue fraud prevention rules	43
10.6 Engagement scoring model, and its use in follow conversion	43
10.7 Notification governance	43
10.8 Geofencing & arrival confirmation	44
10.9 Content moderation	44
10.10 Data retention & privacy baseline	44
10.11 Subscription plans & trial (<i>new, v1.4</i>)	44
10.12 Founding Venue Program & Promo Codes (<i>new, v1.4</i>)	45
10.13 Checkout & platform service fee (<i>new, v1.4</i>)	45
10.14 Admin portal security (<i>new, v1.4</i>)	46
10.15 Delegated venue seat (Venue Team Member / Social Content Creator) (<i>new, v1.5</i>)	46
10.16 Trial & re-claim abuse prevention (Google Place ID) (<i>new, v1.7</i>)	47
11. State Diagrams	48
11.1 Post lifecycle	48
11.2 Follow lifecycle	48

11.3 Venue account / claim verification	48
11.4 Push notification lifecycle	49
11.5 Subscription lifecycle (<i>new, v1.4</i>)	49
12. Sequence Diagrams	50
12.1 Venue claim & onboarding	50
12.2 Post creation & publish	50
12.3 Push notification delivery (discovery + bundling)	51
12.4 QR scan → geofence resolution → follow/arrival	51
12.5 Arrival confirmation on PWA open	52
12.6 Subscription checkout & Paddle payment (<i>new, v1.4</i>)	53
12.7 Admin portal login (MFA) (<i>new, v1.4</i>)	53
13. API Contracts	55
13.1 Auth & accounts	55
13.2 Venues	55
13.3 Posts & media	56
13.4 Discovery	57
13.5 Follow	57
13.6 QR & geofence	57
13.7 Devices & push	57
13.8 Metrics	58
13.9 Static pages & admin	58
13.10 Subscriptions & billing (<i>new, v1.4</i>)	58
13.11 Admin account & content management (<i>new, v1.4</i>)	59
14. Data Model	60
14.1 Entity relationship diagram	60
14.2 Core tables	60
14.3 Analytics event stream (append-only, drives §17)	66
15. Notification Engine	67
15.1 Trigger types	67
15.2 Eligibility computation (per trigger cycle)	67
15.3 Bundling algorithm	67
15.4 Device protection caps (enforced in this order)	68
15.5 Multi-day / recurring post handling	68
15.6 48-hour trial-follow notification scope	68
15.7 Delivery infrastructure	68
15.8 Venue-owner operational pushes	68
15.9 Subscription lifecycle notifications (<i>new, v1.4</i>)	69
16. QR Flow	70
16.1 Issuance	70
16.2 Scan resolution logic	70
16.3 Metrics captured per scan	71
16.4 Edge cases	71
17. Analytics	72
17.1 Validation metrics (network-level, tied to the MVP hypotheses in §4.1)	72
17.2 Post Performance Card (venue-facing, per post)	72
17.3 Venue dashboard (5 visuals max, per source brief's explicit cap)	72
17.4 Admin (platform) dashboard	73
17.5 Refresh cadence & access	73

18. Security	74
18.1 Authentication & session management	74
18.2 Role-based access control (RBAC)	74
18.3 Bot & traffic protection	75
18.4 Application security baseline	75
18.5 Push abuse prevention	75
18.6 Privacy & compliance	75
18.7 Reporting & incident handling	75
19. Performance & Non-Functional Requirements	76
19.1 Scale targets	76
19.2 Response time & availability budgets	76
19.3 Media performance (recap, authoritative detail in §7.5)	76
19.4 Caching & delivery architecture	77
19.5 Database performance	77
19.6 Non-functional requirements	77
20. Deployment	78
20.1 Final MVP stack	78
20.2 Domain structure	78
20.3 Environments & CI/CD	79
20.4 Monitoring, logging, and alerting	79
20.5 Backup & disaster recovery	79
20.6 Rollout plan	79
20.7 Forward-compatibility guardrails (non-blocking for MVP, binding on architecture choices)	79
21. Subscription & Billing (<i>new chapter, v1.4</i>)	80
21.1 Purpose & validation link	80
21.2 Subscription state model	80
21.3 Scheduled jobs	81
21.4 Checkout, Paddle integration, and concurrency	81
21.5 Auditability	81
21.6 Relationship to Admin	81
Appendix A — Decisions Log	82

How to Read This Document

Requirements use RFC 2119-style keywords:

- **shall / must** — mandatory. The build is non-conformant without it.
- **should** — strongly recommended; deviation requires a documented reason.
- **may** — optional/permitted.

Every functional requirement carries a stable ID (e.g., REQ-POST-014) for traceability between this document, tickets, and test cases. Where the source concept brief was ambiguous, contradictory, or silent, this document makes an explicit **PM Decision** and calls it out in a callout block so the assumption is visible and can be challenged before build starts. Nothing in this document narrows or expands MVP scope beyond what the source brief described — decisions resolve *how* a stated requirement behaves, not *whether* it exists.

PM Decision blocks like this one flag a judgment call made to remove ambiguity.

Version 1.4 changelog: this revision incorporated the *Nearby Vibes Additional Requirements — 23 July* brief and added two previously out-of-scope/underspecified capability areas to the MVP: (1) **Subscription & Billing** — a 14-day free trial, Free/Premium plans, a Founding Venue Program and promo-code mechanism, and Paddle.com-processed checkout with lifecycle notifications (§7.12, §21, and updates to §2, §4, §10, §11, §12, §13, §14, §15, §18, §20); and (2) an **expanded Admin Panel** — a separately routed, MFA-protected admin portal, full account/venue/post management, an NYC-venue onboarding alert, and a linked content-report moderation workflow (updates to §7.8, §9.1, §13, §14, §18, §20). This superseded the v1.0–v1.3 framing of subscriptions as a deferred, second-iteration feature (see Appendix A, items 6–7).

Version 1.5 changelog: this revision added a **Delegated Venue Seat** capability — a venue owner may invite, by email in a GitHub-style flow, exactly one team member or content creator per venue account, selecting a role type (**Venue Team Member** or **Social Content Creator**) and a custom seat title before sending the invite. The invited person creates their own account, may create posts and edit/delete only the posts they personally created, and — expanded from the prior Social Media Manager restriction — can view the venue’s overall performance dashboard in addition to their own post detail (§5.3, §6.2–§6.3, §7.1 REQ-ACC-030–038, §10.15, §9.1, §9.7, §14.2, §17.3, §18.2; see Appendix A, item 8).

Version 1.6 changelog: Product confirmed the open item from v1.5 (Appendix A, item 8): Venue Team Member and Social Content Creator share one identical privilege set, with no functional divergence between the two role types. This revision also added a new constraint raised alongside that confirmation: the delegated seat itself is a **Premium-tier capability** — available while a venue is in its 14-day Trial (which carries full Premium access, REQ-SUB-002) or on an active paid Premium subscription, but **not on the Free plan**. If a venue’s subscription lapses to Free (trial expiry without upgrade, or non-renewal), any existing delegated seat is suspended — not deleted — and reactivates automatically if the venue returns to Premium (REQ-ACC-039–040, §10.15, §9.1, §9.7, §11.5, §17.3; see Appendix A, item 9).

Version 1.7 changelog: Product confirmed the suspend-and-restore behavior from v1.6 (Appendix A, item 9) with no changes — that item is now fully closed. This revision added a **trial and re-claim abuse prevention** rule targeting a specific gaming pattern: a venue owner exhausting the 14-day free trial, deleting the account, and re-claiming the same venue under a new email to draw another free trial. Two linked constraints closed this: (1) a given Google Place ID is entitled to exactly **one 14-day free trial, ever** — not once per account, once per Place ID, persisted independent

of any account deletion; and (2) a Google Place ID whose claim was deleted cannot be re-claimed by *any* account for a configurable cooldown period (default **90 days / ~3 months**), with an Admin override for legitimate cases such as a genuine ownership transfer (REQ-ONB-023–026, REQ-SUB-021, §10.16, §10.5, §11.3, §13.2, §14.2; see Appendix A, item 10).

Version 1.8 changelog: Product confirmed both open sub-points from v1.7 (Appendix A, item 10) with no changes to behavior: the one-time trial restriction is **permanent per venue location** (Google Place ID) and never resets, and the 90-day cooldown governs **only the ability to re-claim** that Place ID — the two are confirmed as separate, correctly-scoped constraints exactly as implemented in §10.16. Item 10 is now fully closed; no further open items remain in this document as of v1.8.

1. Executive Summary

Nearby Vibes is a location-based Progressive Web App that lets hospitality venues (bars, restaurants, and venues serving both) publish time-boxed food and drink specials to people physically nearby, and lets those people discover, navigate to, and redeem those specials in the moment. The MVP's job is not to be a complete social platform — it is to prove two things with real usage data: that people within roughly 0.75 miles of a venue will engage with a PWA-delivered special and act on it, and that a QR-code-plus-geofence loop can verify that a post actually converted into a venue visit.

The MVP scope is deliberately narrow: venues post, users discover and follow, the system tracks the path from impression to confirmed arrival, and venues see that path summarized on a simple dashboard. There is no in-app booking and no user-to-user social graph or venue advertising marketplace in this release — those remain explicitly out of scope. **Unlike v1.0–v1.3 of this document, subscriptions and billing via Paddle.com are now in scope for this MVP** (§4.2, §7.12, §21) — venues progress through a 14-day free trial into either a Free plan or a paid Premium plan, with a Founding Venue Program and promo-code mechanism to drive early adoption in Manhattan.

Three platform actors carry the MVP: **Venues**, who create and manage specials, manage their subscription, and see performance data; **Users**, who discover specials nearby, follow venues, and navigate to them; and **Admin**, Nearby Vibes' internal operations staff, who now access a dedicated, MFA-protected portal to manage accounts, moderate content, and monitor onboarding and subscription health. A fourth, narrower actor — the Social Media Manager — is a delegated, restricted venue-side role for agencies/influencers who post on a venue's behalf without full account or billing access.

Success for the MVP is measured by whether nearby users open and act on posts (impressions → opens → navigation clicks → confirmed arrivals), whether venues can see, in a dashboard they trust, that posting a special measurably increases foot traffic during slow trading periods, and — newly in scope — whether the trial-to-paid subscription funnel converts venues into paying Premium customers.

2. Product Vision

Vision statement: Nearby Vibes becomes the fastest way for a hospitality venue to turn a quiet hour into a full one — by putting a real-time, geographically relevant offer in front of the exact people close enough to act on it right now, and proving that it worked.

Long-term direction (post-MVP, non-binding on this build): the PWA architecture shall be designed so that a native mobile app and richer user-tracking/attribution can be added without a platform rewrite.

PM Decision (updated, v1.4): v1.0–v1.3 of this document treated a “subscription/billing layer (Paddle.com)” as a *future* capability the architecture should merely not foreclose. The 23 July Additional Requirements brief moves subscription and billing into the MVP itself. This paragraph, §4.2/§4.3, and the new §7.12/§21 supersede that earlier framing. The forward-compatibility guardrail that remains *genuinely* deferred is the **native mobile app** — the MVP shall not build it, but shall not make architectural choices that foreclose it (§20.7).

The problem this replaces: venues currently buy geographically-blunt, attribution-blind ads (e.g., Google Ads / social boosting) and have no reliable signal that ad spend converted to a customer walking in the door. Nearby Vibes’ differentiation is the closed loop: post → nearby impression → navigation → geofenced arrival confirmation → attributed metric on the venue’s own dashboard — and, newly, a direct incentive (the analytics behind that loop) to convert a free trial into a paid subscription.

Value propositions:

Audience	Value proposition
User	“Show me the best deals and what’s happening around me right now.”
Venue	“Help me fill empty seats during quiet periods — and prove it worked.”

3. Product Principles

These principles shall guide every scoping and design trade-off decision made during the MVP build. Where a feature request conflicts with a principle below, the principle wins unless a PM explicitly overrides it in writing.

1. **Proximity is the product.** Every feature shall reinforce the 0.75 mi / 1.2 km relevance radius. Nothing in the MVP shall encourage discovery of venues outside actionable walking/short-trip distance (the only sanctioned exception is the 48-hour trial-follow city-boundary allowance, which exists to keep an already-engaged user informed, not to broaden discovery).
2. **Friction kills conversion.** Account creation, post creation, post viewing, and — newly — subscription checkout shall each be reducible to the fewest possible steps. Age-gating, consent banners, and PWA-install prompts shall be designed to inform without blocking legitimate use.
3. **Cheap to run, cheap to scale.** Every media, hosting, and notification decision shall default to the lowest-cost option that meets the performance bar in §19, because MVP validation depends on the unit economics being provable at small scale before they're tested at large scale. This now includes the platform service fee model in §10.13, which is designed to make Paddle processing costs self-funding rather than absorbed.
4. **Prove the loop, don't just log the click.** Metrics infrastructure shall prioritize the full attribution chain (impression → open → navigation → geofenced arrival) over vanity counts. A metric that cannot be tied back to a specific post and a specific venue is lower priority than one that can.
5. **Respect the platforms.** The PWA shall behave like a well-mannered guest on iOS/Android/desktop browsers: no background location tracking, no notification spam (see the hard caps in §14 Notification Engine), and no dark patterns in consent, install, or subscription-upgrade prompts.
6. **Design for the app that comes next.** Data models, auth, and tracking shall be built so that a native app can plug into the same backend later without a schema migration that breaks history (see §20.7).

4. MVP Definition

4.1 What the MVP validates

The MVP exists to answer three validation questions (the third added in v1.4), and the product shall be built to generate clean, attributable data against all of them:

1. **Do users within 0.75 mi / 1.2 km engage with the PWA?** Measured via the funnel: post impression → post open → navigation click → geofenced arrival confirmation.
2. **Does the QR-code-plus-geofence loop work as a verification and follow-acquisition mechanism?** Measured via QR scan volume, scan-to-follow conversion, and scan-to-confirmed-visit conversion.
3. **Does the trial-to-paid subscription mechanism convert venues?** (*New, v1.4.*) Measured via trial-start rate, trial-to-Premium conversion rate, Founding Venue Program uptake, and promo-code redemption rate (§10.11–§10.13, §17.4).

4.2 In scope for MVP

- Venue and personal user account creation, including the delegated Social Media Manager role.
- Google Places-backed venue claiming and profile setup.
- Post (special) creation, scheduling, publishing, and archival for food and drink specials.
- Tile view and map view discovery, within a 0.75 mi / 1.2 km default radius.
- Age-gating for alcohol specials (self-attested).
- Follow system: 48-hour trial follow, engagement-based conversion to a full follow.
- Push notifications: discovery pushes, venue-follow pushes, bundling, quiet hours, frequency caps.
- Native-app hand-off navigation (Google Maps / Apple Maps deep links).
- QR-code venue verification, tied to geofence validation.
- Arrival confirmation via geofence + time-window logic.
- **Venue subscription & billing** (*new, v1.4*): a 14-day free trial (no credit card required), automatic downgrade to a Free plan at trial end absent a paid upgrade, a Premium plan (\$39/month or \$390/year) unlocking the analytics dashboard and related metrics, a Founding Venue Program and promo-code mechanism for qualifying NYC venues, Paddle.com-processed checkout with a platform service fee, and pre-expiry/renewal lifecycle notifications (§7.12, §21).
- Venue analytics dashboard and an internal admin/platform dashboard.
- **Expanded Admin Panel** (*new, v1.4*): a dedicated, MFA-protected admin authentication portal at a non-public URL; full view/edit/delete account management for Users and Venues; full view/edit/delete management of Posts; an alert when a new NYC-located venue completes onboarding; and a linked report queue for legal/content moderation with direct view/edit/delete actions (§7.8, §18).
- Static/legal pages (About, For Users, For Businesses, Terms, Privacy, Acceptable Use, Contact).
- Cookie consent, accessibility (WCAG 2.1), and baseline security controls.
- SEO-structured, indexable post and venue pages.

4.3 Explicitly out of scope for MVP

The following are deferred to a subsequent iteration and shall **not** be built now. Where the source brief mentioned a future direction, it is recorded here to prevent scope creep, not to authorize early work:

- In-app/in-PWA turn-by-turn routing (navigation is a hand-off to native maps apps only).
- Native mobile app (iOS/Android) — PWA only.
- User-to-user social graph (comments, follows between users, DMs).
- Venue-side paid promotion/advertising marketplace.
- Automated ID-based age verification (MVP uses self-attestation only).
- In-app video/photo editing beyond crop/reposition/zoom/hero-select (see §9).
- Multi-language / i18n (MVP ships US-English content; structure should not preclude it later).
- Live Photo processing on iOS (explicitly noted in the source brief as “optional later”).
- In-app subscription cancellation self-service beyond the basic cancel action in §21.4 (full dunning/win-back flows, multi-tier plans beyond Free/Premium, and seat-based pricing are deferred).

PM Decision (superseded item, v1.4): v1.0–v1.3 listed “Subscriptions, billing, or any Paddle.com integration” here as out of scope. The 23 July Additional Requirements brief brings this in scope for the MVP. That bullet is removed from this list; see §4.2, §7.12, and §21 for the resulting requirements, and Appendix A item 6 for the full audit trail of this reversal.

PM Decision: Where the source brief included forward-looking infrastructure notes (Firebase Auth/FCM “kept” for future improved tracking, PWA “needing to move to an app”) these are treated as non-functional architectural constraints (§19, §20), not MVP features. They inform technology choice, not MVP scope.

4.4 Target launch parameters

Parameter	Value
Launch city	Manhattan, NY
Minimum supported scale	10,000+ users, 200+ venues
Relevance radius (default discovery)	0.75 mi / 1.2 km
Venue geofence radius	Property line + 30 m buffer
Platform	Progressive Web App (installable, Safari + Chrome compatible)
Founding Venue Program cap	First 50 Manhattan venues (configurable)

5. User Personas

5.1 Maya — The Nearby User

- **Age:** 27. Lives and works in Manhattan.
- **Goal:** Decide what to do in the next 30–60 minutes based on what’s actually good and close by, not what an algorithm decided to show her from three miles away.
- **Behavior:** Checks her phone while walking; will not fill out a form or read a paragraph to find a deal. Wants a blurred cocktail tile to be intriguing, not an obstacle.
- **Needs from Nearby Vibes:** instant tile/map view of what’s live nearby, one-tap navigation, minimal onboarding friction, and confidence that the “8 minutes left” timer is real.
- **Frustration it solves:** Instagram/social discovery is not proximity-aware or time-boxed; Google Maps shows venues but not live offers.

5.2 Sam — The Venue Owner/Manager

- **Role:** Owner-operator or GM of a bar/restaurant with 40–120 covers.
- **Goal:** Fill seats during a predictably slow window (e.g., Tuesday 4–6pm) without paying for an ad campaign that can’t prove it worked — and to do so cheaply while the business is proving itself out.
- **Behavior:** Time-poor. Wants to post a special in under two minutes from a phone, wants one glaring number (“Customers Brought In”) that justifies the ten minutes it took, and wants to try the product before paying for it.
- **Needs from Nearby Vibes:** fast post creation, a trustworthy claimed-venue profile, a follower list that grows without manual work, a dashboard that doesn’t require analytics literacy to read, and — newly — a no-risk 14-day trial with a clear, fairly priced upgrade path (\$7.12).
- **Frustration it solves:** Google/Meta ad spend with no attendance attribution.

5.3 Jordan — The Delegated Venue Seat (Venue Team Member / Social Content Creator)

PM Decision (v1.5): the source brief introduces this persona as a named seat a venue owner invites, choosing one of two role types — **Venue Team Member** or **Social Content Creator** — and assigning a custom title (e.g., “Marketing Coordinator”). This document treats “Social Media Manager,” used in earlier revisions and still present in some requirement IDs (REQ-ACC-03x) and screen names for continuity, as the generic engineering term for this same delegated seat. The two selectable role types share one privilege set (§7.1) — the distinction is organizational/labeling for the venue owner, not a difference in system permissions, unless Product specifies otherwise.

- **Role:** A junior team member or an outside social content creator, invited by the venue owner to handle the venue’s day-to-day posting. Typically not the account holder — often the person actually running the venue’s social presence.
- **Goal:** Post specials on behalf of a venue quickly, without needing (or wanting) access to the venue’s account settings, billing, follower list, or other sensitive account areas.

- **Behavior:** Creates their own login (separate from the owner's); works within a single venue account (each venue account may fill exactly one such seat, §10.15); wants to see how their own content is performing in the context of how the venue overall is doing.
- **Needs from Nearby Vibes:** an invite-based account, issued by email with a GitHub-style role picker at invite time, with a permission set scoped to their own posts (create, edit, delete only posts they created) and dashboard access that includes both the venue's overall performance and the detail for their own posts (§7.1, §10.15) — but not billing, account settings, or other people's post detail.

5.4 Alex — The Platform Admin (internal, Nearby Vibes staff)

- **Role:** Internal operator responsible for trust & safety, platform health, and — newly — account/subscription oversight.
- **Goal:** Catch fraudulent venue claims, respond to reported content quickly, see network-wide health without digging through raw data, and manage the full lifecycle of user/venue accounts and posts through one secure, purpose-built portal rather than ad hoc database access.
- **Needs from Nearby Vibes:** a dedicated, MFA-protected admin portal at a non-discoverable URL (§7.8, §18); admin dashboard (§17); report queue with linked view/edit/delete takedown capability (§7.8); fraud-prevention signals surfaced automatically (duplicate Place ID claims, multi-venue-per-account attempts); full account management for Users and Venues; and an alert whenever a new, fully onboarded venue located in New York City joins the network.

6. User Stories

Stories are grouped by persona and mapped to the functional requirement IDs that satisfy them (see §7). Each story follows the standard “As a / I want / so that” form with an explicit acceptance pointer.

6.1 Maya (User) stories

ID	Story	Satisfied by
US-01	As a nearby user, I want to see live specials within walking distance the moment I open the app, so I don't have to search for what's around me.	REQ-UX-001-006
US-02	As a user under the drinking age (or who hasn't confirmed), I want to still see food specials without being blocked by an age gate, so the app stays useful to me.	REQ-BIZ-001-003
US-03	As a user, I want a one-tap “Navigate” button that opens my phone's native maps app, so I don't have to copy an address.	REQ-NAV-001-002
US-04	As a user, I want to follow a venue so I get notified the next time they post, without being spammed by every venue in the city.	REQ-PUSH-005-012
US-05	As a user without location permission granted, I want to understand why the app needs it and still see something useful (fuzzy results) rather than a blank screen.	REQ-UX-020-021
US-06	As a user, I want to scan a venue's QR code and immediately see their live special, so I don't have to search for the venue in-app while standing there.	REQ-QR-001-006

ID	Story	Satisfied by
US-07	As a user, I want to be prompted (not forced) to install the app to my home screen after I've shown I'm interested, so I get notifications going forward.	REQ-UX-030-033

6.2 Sam (Venue) stories

ID	Story	Satisfied by
VS-01	As a venue owner, I want to claim my venue using my existing Google Maps listing, so I don't have to re-enter my name, address, and hours.	REQ-ONB-010-018
VS-02	As a venue owner, I want to create and publish a drink or food special in under two minutes from my phone, so posting doesn't feel like a chore.	REQ-POST-001-020
VS-03	As a venue owner, I want to see exactly how many people my post reached, opened, navigated from, and actually visited, so I can decide whether to post again.	REQ-METRICS-010-025
VS-04	As a venue owner, I want to invite one team member or content creator, by email, to post on my behalf, choosing their role type and a title before I send the invite, without giving them access to my account settings or billing.	REQ-ACC-030-038
VS-05	As a venue owner, I want to be reminded automatically when my special has ended, so I remember to post the next one and don't leave a dead post live.	REQ-PUSH-020-021
VS-06	As a venue owner posting a drink special, I want the legal disclaimer added automatically, so I don't have to remember compliance language every time.	REQ-POST-025
VS-07	<i>(New)</i> As a venue owner, I want a free 14-day trial with no credit card required, so I can prove the product works before I pay for it.	REQ-SUB-001-004
VS-08	<i>(New)</i> As a venue owner, I want to see and act on pricing at any time from the site footer, not just when my trial is ending, so I can upgrade on my own schedule.	REQ-SUB-005, REQ-SUB-011

ID	Story	Satisfied by
VS-09	(New) As a qualifying Manhattan venue, I want to see a clear, time-limited Founding Venue offer, so I understand exactly why signing up early is worth it.	REQ-SUB-009–010
VS-10	(New) As a venue owner, I want to know exactly what I’m being charged — plan price plus the processing fee — before I confirm payment, so there are no surprises at checkout.	REQ-SUB-012–013
VS-11	(New) As a venue owner, I want to be warned before my subscription lapses, so I don’t lose analytics access without warning.	REQ-SUB-016–017

6.3 Jordan (Delegated Venue Seat) stories

ID	Story	Satisfied by
SM-01	As an invited team member or content creator, I want my own login separate from the venue owner’s, so I can work on the venue account safely using my own details.	REQ-ACC-031
SM-02	As an invited team member or content creator, I want to create posts and edit or delete the ones I created, so I can do my job without needing owner-level access.	REQ-ACC-032
SM-03	(New) As an invited team member or content creator, I want to see the venue’s overall performance alongside the detail for my own posts, so I understand how my work contributes to the bigger picture — without seeing other creators’ individual post detail, account settings, or billing.	REQ-ACC-034, REQ-METRICS-024, REQ-SUB-020

6.4 Alex (Admin) stories

ID	Story	Satisfied by
AD-01	As an admin, I want to be notified the moment a post or venue is reported, so I can review and remove it quickly.	REQ-POST-030–032, REQ-ADMIN-018–019

ID	Story	Satisfied by
AD-02	As an admin, I want automated blocking of duplicate Google Place ID claims, so venues can't be squatted or duplicated.	REQ-ONB-016–018
AD-03	As an admin, I want a network-level dashboard (installs, permission opt-in rates, QR performance, follower growth, and now subscription health) so I can see platform health at a glance.	REQ-METRICS-030–036, §17.4
AD-04	<i>(New)</i> As an admin, I want to log into a separate, non-public admin portal protected by a password and an authenticator app, so the operations console isn't a soft target for credential attacks.	REQ-ADMIN-010–013
AD-05	<i>(New)</i> As an admin, I want full view/edit/delete control over any user account, venue account, or post, so I can act decisively on fraud, abuse, or support requests without engineering involvement.	REQ-ADMIN-014–016
AD-06	<i>(New)</i> As an admin, I want to be notified when a new, fully onboarded venue located in New York City joins the platform, so I can track our core-market growth directly.	REQ-ADMIN-017
AD-07	<i>(New)</i> As an admin, when a report comes in, I want a direct link straight to the content in question with view/edit/delete actions right there, so I don't have to hunt for it separately.	REQ-ADMIN-018–019

7. Functional Requirements

This is the master requirement catalog. Detailed mechanics for notifications, QR, analytics, and — new in v1.4 — subscriptions are elaborated in their dedicated chapters (§15, §16, §17, §21); this catalog is the authoritative ID reference for all of them. All requirements below are P0 (MVP-blocking) unless marked P1 (should-have, non-blocking).

7.1 Accounts & Authentication (ACC)

ID	Requirement
REQ-ACC-001	The system shall let a visitor create an account by first selecting an account type: Personal or Business.
REQ-ACC-002	For Personal accounts, the system shall support account creation via Google sign-in, Apple sign-in, or email/password.
REQ-ACC-003	For Business accounts, the system shall support account creation via Google sign-in or a business email address.
REQ-ACC-004	The system shall reject Business account creation when the supplied email domain matches the blocked free-email domain list (§10.4).
REQ-ACC-005	The system shall require acceptance of the Terms of Service and Privacy Policy before either account type can complete registration.
REQ-ACC-006	The system shall enforce a password policy requiring a minimum of 10 characters, at least one uppercase letter, one number, and one symbol, for email/password accounts.
REQ-ACC-007	The system shall provide a “remain signed in” option, capped at a 30-day session lifetime, after which re-authentication is required.
REQ-ACC-008	The system shall provide a forgot-password flow that sends a reset link to the account’s verified email address only.
REQ-ACC-009	The system shall allow the user to set a new password after following a valid, non-expired reset link, and shall expire reset links after 60 minutes.
REQ-ACC-010	The system shall send a verification email upon signup and shall block onboarding completion until the email is verified.
REQ-ACC-011	The system shall apply Cloudflare Turnstile to all account-creation and login forms as bot protection.

ID	Requirement
REQ-ACC-012	The system shall apply Cloudflare Bot Management and Cloudflare Rate Limiting to all public-facing traffic.
REQ-ACC-030	A Venue account owner shall be able to invite one delegated team member (“Social Media Manager” seat, §5.3) by email, in a GitHub-style invite flow.
REQ-ACC-031	An invited delegated team member shall create their own login credentials, using their own details, distinct from the venue owner’s, before gaining access.
REQ-ACC-032	A delegated team member account shall be permitted to create posts for the venue(s) they were invited to, and shall be permitted to edit or delete only the posts they personally created.
REQ-ACC-033	A delegated team member account shall be denied access to venue settings, follower data, other creators’ individual post detail beyond the venue-wide dashboard (REQ-ACC-034), subscription/billing screens, and account deletion/edit functions.
REQ-ACC-034	A delegated team member shall be able to view the venue’s overall performance dashboard (§17.3) <i>and</i> detailed performance metrics (push sent, push opens, views, arrivals) scoped to posts they personally created; they shall not see other creators’ individual post-level detail.
REQ-ACC-035	<i>(New, v1.5)</i> A Venue account shall be permitted exactly one active delegated team member seat at a time; the system shall reject a second concurrent invite while one is pending or accepted, and shall display that the seat is already filled.
REQ-ACC-036	<i>(New, v1.5)</i> Prior to sending the invite, the venue owner shall select a role type for the seat: Venue Team Member or Social Content Creator . Both role types share the identical privilege set in REQ-ACC-032–034 (§10.15); the distinction is a label for the venue owner’s own organization, not a difference in system permissions.
REQ-ACC-037	<i>(New, v1.5)</i> The venue owner shall be able to set a free-text title for the seat (e.g., “Marketing Coordinator”), displayed alongside the invited person’s name wherever their identity appears (post byline, admin views, dashboard attribution).
REQ-ACC-038	<i>(New, v1.5)</i> If a venue’s delegated seat is revoked, the venue owner shall be able to issue a new invite (selecting a role type and title again) to fill the now-vacant seat.
REQ-ACC-039	<i>(New, v1.6)</i> The system shall permit a venue owner to send a delegated-seat invite (REQ-ACC-030) only while the venue’s subscription state is Trial or an active paid Premium plan (§21.2); the system shall block the invite action and explain the Premium requirement while the venue is on the Free plan.

ID	Requirement
REQ-ACC-040	<i>(New, v1.6)</i> If a venue's subscription transitions to the Free plan while a delegated seat is Pending or Active, the system shall suspend that seat's access (block login-gated post creation/edit/delete and dashboard access) without deleting the seat record, its role type, or its title; the system shall automatically restore full seat access, with no re-invite required, if the venue's subscription returns to Trial or Premium.

7.2 Venue Onboarding & Claiming (ONB)

ID	Requirement
REQ-ONB-010	The system shall let a Business account search for its venue via a Google Places Autocomplete integration, restricted to place categories indicating food service, alcohol service, or both.
REQ-ONB-011	The system shall import venue name, address, opening hours, and venue type from the selected Google Places result.
REQ-ONB-012	The system shall display the selected venue on a map for visual confirmation before the venue proceeds.
REQ-ONB-013	The system shall allow the venue to confirm the match or return to search if incorrect.
REQ-ONB-014	The system shall enforce a 1:1 relationship between a Google Place ID and a Nearby Vibes venue account.
REQ-ONB-015	The system shall reject a claim attempt against a Place ID already claimed by another account and shall display: "This venue is already registered."
REQ-ONB-016	The system shall reject venue account creation when the requesting account has already created another venue (one venue per business account in MVP).
REQ-ONB-017	The system shall flag for admin review any claim attempt that trips fraud-prevention rules (§10.5) rather than silently succeeding.
REQ-ONB-018	The system shall require a venue to complete its profile (§7.2 fields below) before its first post can be published.
REQ-ONB-020	The venue profile shall capture: logo image, background/showcase image, venue name and address (prepopulated, editable only via re-claim), category type, a multi-select feature list (e.g., sports bar, dance floor, beer garden, restaurant), map location, and a system-generated unique venue ID.
REQ-ONB-021	The venue profile shall include a "Report" action that notifies the Admin persona with a link to the reported venue.

ID	Requirement
REQ-ONB-022	(New) Upon a venue completing all onboarding tasks (profile complete, REQ-ONB-018) and enrolling in the trial (REQ-SUB-001), the system shall evaluate whether the claimed venue's address resolves within New York City, NY, and — if so — shall raise the Admin onboarding alert defined in REQ-ADMIN-017.
REQ-ONB-023	(New, v1.7) The system shall retain a claim history record for every Google Place ID ever claimed on Nearby Vibes, keyed by <code>google_place_id</code> , that persists independent of the lifecycle of any venue account associated with it — the record shall not be deleted when the associated venue account is deleted (§10.16, §14.2).
REQ-ONB-024	(New, v1.7) The system shall reject a claim attempt against a Google Place ID whose most recently associated venue account was deleted less than a configurable cooldown period ago (default: 90 days / ~3 months), and shall display that the venue cannot be re-claimed yet (§10.16).
REQ-ONB-025	(New, v1.7) The system shall flag any claim attempt rejected under REQ-ONB-024 to the Admin fraud queue (§10.5), distinct from — and in addition to — the existing duplicate-active-claim rejection in REQ-ONB-015.
REQ-ONB-026	(New, v1.7) An Admin shall be able to override the cooldown in REQ-ONB-024 for a specific Google Place ID (e.g., a confirmed legitimate ownership change), with the override logged to the admin audit trail (REQ-ADMIN-004, REQ-ADMIN-020).

7.3 Discovery Experience — Tile & Map (UX)

ID	Requirement
REQ-UX-001	On first visit, the system shall display the Tile View of currently active specials as the default screen.
REQ-UX-002	The system shall provide a toggle to switch between Tile View and Map View at all times.
REQ-UX-003	Each tile shall display: post hero image, post headline, price or percentage-off, venue name, venue address, a live decaying countdown timer (time remaining + validity window), and distance from the user.
REQ-UX-004	The system shall not provide a text search feature in the MVP; discovery is limited to proximity-based browsing and manual map panning.
REQ-UX-005	The system shall let a user pan the map to preview specials around a location other than their current position.
REQ-UX-006	Tile View and Map View shall both default to the 0.75 mi / 1.2 km relevance radius from the user's current (or panned) location.

ID	Requirement
REQ-UX-010	Map View shall render venues within the default radius and shall let the user zoom out beyond it.
REQ-UX-011	Map View shall display a banner showing the live count of specials currently within the visible radius.
REQ-UX-012	Map View shall use Mapbox GL JS as the MVP mapping provider, for its native geofence/polygon rendering and custom marker-icon support (confirmed, §20.1).
REQ-UX-013	Map View shall cluster markers when venue density would otherwise cause overlap.
REQ-UX-014	Map markers shall render in three states: active drink special (highlighted cocktail icon), active food special (highlighted food icon), and no active post (neutral/greyed marker, still visible to communicate ecosystem size).
REQ-UX-015	Tapping a marker once shall open a compact preview tile (cover image, title, venue, address, time remaining); tapping again shall open the full post page.
REQ-UX-016	Map View shall provide street and satellite layer toggles and a “you are here” indicator.
REQ-UX-017	Preview cards on the map shall be compact by design; the system shall not render oversized preview cards that obscure the map.
REQ-UX-020	If a user has not granted location permission, the system shall return fuzzy, city/neighborhood-level results rather than an empty state.
REQ-UX-021	If location permission is not granted, the system shall display a dismissible prompt explaining that precise location improves result accuracy, with a control to grant it.
REQ-UX-030	The system shall display a bottom-banner PWA install prompt after 30 seconds of session time on a non-installed, supported browser.
REQ-UX-031	The install prompt shall not display if <code>display-mode: standalone</code> is detected (already installed) or if the user dismissed it within the last 7 days.
REQ-UX-032	The install prompt shall show at most once per 7-day period per device.
REQ-UX-033	On acceptance, the system shall display device-specific (iOS vs. Android) step-by-step install instructions.
REQ-UX-040	The system shall display a cookie consent banner on first visit with Accept, Reject, and Manage Preferences controls, meeting US/CCPA and EU/GDPR requirements; essential/functional cookies shall remain active regardless of choice.
REQ-UX-041	The footer shall display the current copyright line, links to Terms of Service, Privacy Policy, and Acceptable Use Policy, a location-access toggle, a manual “install app” trigger, and — <i>new</i> — a Pricing link (REQ-SUB-005).
REQ-UX-042	The PWA shall conform to WCAG 2.1 (minimum Level AA) across all user-facing screens.

7.4 Post / Special Lifecycle (POST)

ID	Requirement
REQ-POST-001	The system shall support two post types: Drink Special and Food Special.
REQ-POST-002	A post shall move through exactly four states: Draft → Scheduled → Active → Archived (see §11 State Diagrams).
REQ-POST-003	The system shall enforce a maximum of one Active post per venue at any given time.
REQ-POST-004	The system shall enforce a maximum post duration of 7 days from first activation to final archival.
REQ-POST-005	The system shall support recurring daily time windows within a single post's date range (e.g., 2–4pm daily for 7 days); the post shall render as active to users only during those windows.
REQ-POST-006	The system shall auto-transition a post from Active to Archived at the end of its final time window and shall stop public display immediately at expiry.
REQ-POST-007	Draft and Archived posts shall be duplicable, with the system generating a new unique post ID for the duplicate.
REQ-POST-008	Archived posts shall not be deletable by the venue (<i>the Admin may delete any post per REQ-ADMIN-016</i>).
REQ-POST-009	An Active post shall be editable or manually unpublished (moved to Archived) by the venue before its scheduled end time.
REQ-POST-010	A post shall require these fields to move out of Draft: post type, title (≤60 characters), hook (≤120 characters), price or % off, description (≤500 characters), start date/time, end date/time, hero image, and (for Drink posts) the alcohol compliance disclaimer acknowledgment.
REQ-POST-011	A post may optionally include: conditions text (≤100 characters), a compliance/food-field note (≤100 characters), and one media asset (one image, or one video up to 15 seconds).
REQ-POST-012	The system shall allow exactly one hero image (used on tiles/cards) and, separately, up to one additional media asset (image or ≤15-second video) shown within the full post.
REQ-POST-013	If the post's media asset is a video, the system shall render it as one swipeable slide and shall require a manual tap to play; autoplay is prohibited.
REQ-POST-014	The system shall not render grid layouts or stacked media sections within a post — one hero container only.
REQ-POST-015	The system shall let a venue preview the fully rendered post before scheduling or publishing it.
REQ-POST-016	The system shall auto-generate on publish: a unique special ID, a creation timestamp, and a venue ID linkage.

ID	Requirement
REQ-POST-020	For Drink Special posts, the system shall append the standard alcohol compliance disclaimer (§10.2) automatically and immutably; venues shall not be able to edit or remove it.
REQ-POST-021	For Drink Special posts, the system shall surface an “Alcohol Advertising Help” reference page and a “Tips to Boost Results” reference page during creation.
REQ-POST-022	Upon a post’s transition to Archived, the system shall send a push notification to the venue owner and/or Social Media Manager responsible for that post.
REQ-POST-030	Every rendered post shall display a “Report” action.
REQ-POST-031	Selecting Report shall route to the Contact page and shall notify the Admin persona with a reference to the reported post, appearing in the report queue defined in REQ-ADMIN-018–019.
REQ-POST-032	The Admin shall be able to view reported content and remove it from public display.

7.5 Media Pipeline (MEDIA)

ID	Requirement
REQ-MEDIA-001	The system shall accept photo uploads in JPG/JPEG, PNG, HEIC, and WebP; HEIC shall be auto-converted server-side.
REQ-MEDIA-002	The system shall accept video uploads in MP4, MOV, M4V, and WebM, and shall normalize output to MP4 (H.264).
REQ-MEDIA-003	The primary post canvas shall be 1080×1920 px (9:16 portrait); the system shall provide an in-app crop frame at upload.
REQ-MEDIA-004	The system shall auto-format images per source aspect ratio: 9:16 used directly; 4:5 intelligently expand/crop to 9:16; square center-placed (with optional blurred fill); landscape intelligently cropped.
REQ-MEDIA-005	The system shall apply the same aspect-ratio handling logic to video (9:16 direct; landscape center-crop or blurred vertical fill; square placed within vertical canvas).
REQ-MEDIA-006	The system shall reject video uploads exceeding 15 seconds at the point of upload, with a clear error message.
REQ-MEDIA-007	The system shall enforce a maximum pre-compression upload size of 5MB for images; the equivalent video ceiling is defined in §19 Performance.
REQ-MEDIA-008	The system shall process every uploaded image exactly once at upload time (not per-request), generating thumbnail (300px), feed (800px), and full (1600px max) derivatives in WebP.

ID	Requirement
REQ-MEDIA-009	The system shall store processed media using a predictable, cache-friendly path structure keyed by venue ID and asset role (e.g., <code>/venueId/postId/thumb.webp</code>).
REQ-MEDIA-010	The system shall serve all media through a CDN on a dedicated media subdomain, never through the application server or the Next.js image optimizer.
REQ-MEDIA-011	The system shall provide in-app editing limited to: crop, reposition, zoom, and hero/thumbnail selection — no filters or advanced editing in MVP.
REQ-MEDIA-012	The system shall support upload via native file picker and camera/gallery access on mobile, and drag-and-drop plus multi-file selection on desktop.
REQ-MEDIA-013	The system shall show upload progress and a rendered preview before the venue can proceed to publish.

7.6 Navigation (NAV)

ID	Requirement
REQ-NAV-001	The system shall provide a “Navigate” action on both the post view and the venue profile.
REQ-NAV-002	On tap, the system shall hand off to the device’s native maps application (Apple Maps on iOS, Google Maps on Android) using the venue’s coordinates; the system shall not implement in-app routing.
REQ-NAV-003	The system shall log a <code>navigation_click</code> event, attributed to the source post and venue, at the moment of hand-off (see §17 Analytics).

7.7 Static & Legal Pages (STATIC)

ID	Requirement
REQ-STATIC-001	The system shall provide editable static pages for: About, For Users, For Businesses, Terms of Service, Privacy Policy, Acceptable Use Policy, and Contact.
REQ-STATIC-002	Static pages shall support rich text and embedded YouTube video, editable by an admin without a code deploy.
REQ-STATIC-003	Static page edits shall be version-controlled, with the ability to view or roll back to a prior version.

7.8 Admin (ADMIN)

Expanded substantially in v1.4 — REQ-ADMIN-010 through REQ-ADMIN-020 are new; REQ-ADMIN-001–004 are unchanged from v1.3.

ID	Requirement
REQ-ADMIN-001	The system shall provide an internal admin console for reviewing reported venues/posts and removing content.
REQ-ADMIN-002	The system shall provide the Admin with a network-level performance dashboard (detailed in §17.4), including subscription health metrics (§17.4).
REQ-ADMIN-003	The system shall alert the Admin in near-real-time (push/email) when new content is reported.
REQ-ADMIN-004	The system shall log all admin moderation and account-management actions (removal, restore, edit, delete) with actor, timestamp, target, and reason.
REQ-ADMIN-010	(New) The system shall provide a dedicated Admin authentication portal at a distinct, non-standard URL path that is not linked from any public-facing navigation, footer, sitemap, or robots.txt allow-list, so as to reduce discoverability by automated scanners (§18, §20.2).
REQ-ADMIN-011	(New) Admin portal login shall require a registered business email address, a password meeting the policy in REQ-ACC-006, and a time-based one-time passcode (TOTP) from an authenticator app as a mandatory second factor.
REQ-ADMIN-012	(New) The system shall apply Cloudflare Turnstile, Cloudflare Bot Management, and a stricter rate limit to the Admin portal's login and MFA-verification endpoints than to standard user/venue login (§18.3).
REQ-ADMIN-013	(New) The system shall lock an Admin account for a configurable cool-down period after 5 consecutive failed login or MFA attempts and shall alert on-call staff of the logout.
REQ-ADMIN-014	(New) The system shall let an Admin view, edit, and delete any User account.
REQ-ADMIN-015	(New) The system shall let an Admin view, edit, and delete any Venue account, including its associated posts, subscription state, and Social Media Manager invitations.
REQ-ADMIN-016	(New) The system shall let an Admin view, edit, and delete any individual Post, independent of whether that post has been reported.
REQ-ADMIN-017	(New) The system shall notify the Admin (push notification and an entry in a dedicated onboarding queue) whenever a Venue account completes all onboarding tasks (REQ-ONB-018) and the venue's claimed address resolves within New York City, NY (REQ-ONB-022).
REQ-ADMIN-018	(New) Every report of a Venue profile or a Post for a legal or content matter shall appear as an entry in a simple Admin report list and shall trigger a push notification to the Admin, in addition to the existing behavior in REQ-ADMIN-003.

ID	Requirement
REQ-ADMIN-019	(New) Each report-list entry shall include a direct link to the reported Venue profile or Post, and shall provide view, edit, and delete actions on that content directly from the report entry, without requiring separate navigation to a different admin screen.
REQ-ADMIN-020	(New) All Admin account-management and content-moderation actions (edit, delete, restore) shall be logged to the admin audit trail (REQ-ADMIN-004) with actor, timestamp, target type/ID, and action type.

7.9 SEO (SEO)

ID	Requirement
REQ-SEO-001	Every post page shall render server-side with a unique title, meta description, H1, and a search-optimized first paragraph.
REQ-SEO-002	Post URLs shall follow a human-readable, geo-descriptive slug pattern: nearbyvibes.com/{city}/{category}/{venue-slug}-{special-slug} rather than an opaque numeric ID.
REQ-SEO-003	Every post page shall embed structured data: Offer, LocalBusiness, Event (where applicable), GeoCoordinates, OpeningHours, and PriceSpecification schema.org markup.
REQ-SEO-004	Archived posts shall remain accessible at stable, indexable URLs and shall be marked as ended in both visible copy and structured data, rather than being deleted or 404'd.
REQ-SEO-005	The system shall generate and maintain an XML sitemap and shall support submission to Google Search Console. The Admin portal path (REQ-ADMIN-010) shall be excluded from the sitemap and disallowed in robots.txt.
REQ-SEO-006	The system shall remove expired/archived posts from active sitemap promotion (still indexable, but not prioritized) once past a 90-day staleness threshold. (P1)

7.10 Push Notifications (PUSH)

Full mechanics, bundling, and caps are specified in §15; these are the catalog-level “shall” statements referenced by the user stories in §6.

ID	Requirement
REQ-PUSH-001	The system shall send a General Nearby Push when a special is posted within 0.75mi/1.2km of an eligible device, whether or not that device follows the venue (§15.1).
REQ-PUSH-002	The system shall send a Venue Follow Push to any device with a Trial or Hard follow whenever the followed venue posts, evaluated up to the venue's city-boundary actionability limit (§15.1, §15.6).
REQ-PUSH-003	The system shall consolidate multiple qualifying posts for the same device into a single bundled push within a 15-minute window rather than sending one push per post (§15.3).
REQ-PUSH-004	Tapping a bundled push shall open a filtered Tile View containing only the posts in that bundle (§15.3).
REQ-PUSH-005	The system shall enforce a minimum 60-minute interval between pushes to the same device (§15.4).
REQ-PUSH-006	The system shall cap a device at 2 pushes/bundles within a rolling 24 hours (§15.4).
REQ-PUSH-007	The system shall cap a device at 8 pushes within a rolling 7 days, with the tighter of the 24-hour/7-day caps always winning (§15.4).
REQ-PUSH-008	The system shall cap a venue at 2 push-eligible posts per day, regardless of how many devices are nearby (§15.4).
REQ-PUSH-009	The system shall hold (not drop) any push that would otherwise fire between 12:00 AM–8:00 AM venue-local time, and shall deliver it at 8:00 AM instead (§15.4).
REQ-PUSH-010	The system shall fire the discovery/follow push for a multi-day or recurring post exactly once, at first activation, and shall not re-fire it on subsequent daily windows (§15.5).
REQ-PUSH-011	While a follow is in Trial state, the system shall permit push eligibility outside the 0.75mi radius as long as the device remains within the venue's city boundary or Google Sublocality Level 1 region (§15.6).
REQ-PUSH-012	The system shall deliver all user-facing pushes via Firebase Cloud Messaging Web Push, with the PWA service worker handling receipt and click-through routing (§15.7).
REQ-PUSH-020	The system shall send an operational push to the venue owner/Social Media Manager when their post transitions to Active (§15.8).
REQ-PUSH-021	The system shall send an operational push to the venue owner/Social Media Manager 10 minutes after a post expires, suppressed if a new Active post was created within 9 minutes of that expiry (§15.8).
REQ-PUSH-030	<i>(New)</i> The system shall send subscription-lifecycle operational pushes (and matching emails) to the venue owner per the schedule in §15.9.

7.11 QR (QR)

Full mechanics are specified in §16; these are the catalog-level “shall” statements referenced by the user stories in §6. **Note:** these supersede any earlier per-venue-URL framing — the confirmed design is one universal QR code resolved via geofence, not a venue-specific URL.

ID	Requirement
REQ-QR-001	The system shall issue a single universal QR code (https://nearbyvibes.com/qr) shared by every venue; the code shall not encode a venue identifier (§16.1).
REQ-QR-002	On scan, if location sharing is not granted, the system shall display a prompt explaining location is required for accurate results before revealing any venue-specific content (§16.2).
REQ-QR-003	The system shall resolve a scan’s venue via the geofence layer into exactly one of three tiers — exact match ($\leq 30\text{m}$), nearby fallback ($30\text{--}100\text{m}$), or out of range ($>100\text{m}$) — and shall render tier-appropriate content per §16.2.
REQ-QR-004	When multiple venues’ geofences match a scan within the exact-match tie-break margin, the system shall present a disambiguation list rather than guessing a single venue (§16.2).
REQ-QR-005	The system shall log a <code>qr_scans</code> event on every scan regardless of resolution outcome, including resolved venue (if any), distance tier, and timestamp (§16.3).
REQ-QR-006	If the exact-match-resolved venue has no Active post, the system shall display “No venue live specials right now — check back soon” and route to Home Tile View (§16.2).

7.12 Subscription & Billing (SUB) — *New in v1.4*

Full mechanics, business rules, state/sequence diagrams, and data model are specified in §21 and cross-referenced from §10.11–§10.13, §11.5, §12.6, §13.10, §14.2, and §15.9; this is the catalog-level “shall” reference.

ID	Requirement
REQ-SUB-001	The system shall enroll every newly claimed venue account into a 14-day free trial automatically upon successful venue claim, with no credit card required to start.
REQ-SUB-002	The system shall grant trial venues full Premium-tier feature access for the duration of the 14-day trial.
REQ-SUB-003	The system shall downgrade a venue automatically to the Free plan at the end of the 14-day trial unless the venue has completed a paid subscription checkout.
REQ-SUB-004	The system shall send push and email prompts to the venue owner encouraging upgrade at defined milestones during the trial (§15.9).

ID	Requirement
REQ-SUB-005	The system shall allow a venue to access pricing and initiate a purchase at any time via a “Pricing” link in the global footer (REQ-UX-041), independent of trial state.
REQ-SUB-006	The system shall offer two Premium billing cycles: Monthly (\$39/month) and Annual (\$390/year).
REQ-SUB-007	The Premium plan shall unlock, at minimum: the analytics dashboard, QR scan analytics, follower growth metrics, Soft → Active conversion metrics, post performance, push notification analytics, and venue insights (§17).
REQ-SUB-008	The system shall support a configurable Promo Code capability that grants qualifying venues 50% off either their first 12 months of the annual Premium plan or 12 consecutive monthly Premium billing cycles (§10.12).
REQ-SUB-009	The system shall support a “Founding Venue Program” promotion limited to a configurable cap (default: the first 50 Manhattan venues), displayed on the subscription/pricing panel with the confirmed program copy (§10.12).
REQ-SUB-010	The system shall track Founding Venue Program enrollment count in real time and shall stop offering/displaying the program once the cap is reached.
REQ-SUB-011	The system shall route a venue selecting a paid plan (Promo or standard) to a checkout flow.
REQ-SUB-012	At checkout, the system shall calculate and display a platform service fee equal to 5% of the plan price plus \$0.50 USD, added on top of the plan price, to cover Paddle.com processing costs (§10.13).
REQ-SUB-013	The system shall submit the plan price plus platform service fee as a single total charge to Paddle.com for payment processing.
REQ-SUB-014	On a successful Paddle payment confirmation (webhook), the system shall activate the corresponding subscription (Premium Monthly, Premium Annual, or Promo variant) immediately.
REQ-SUB-015	On a failed or declined payment, the system shall retain the venue’s current plan state and shall display a clear checkout error with a retry option.
REQ-SUB-016	The system shall notify the venue owner via push and email prior to a paid subscription’s renewal/expiry date (§15.9).
REQ-SUB-017	If a paid subscription is not renewed by its expiry date, the system shall downgrade the venue to the Free plan and shall suppress Premium-only features (§17) from that point forward.
REQ-SUB-018	The system shall record every subscription state transition (trial start, trial expiry, upgrade, downgrade, promo applied, renewal, payment failure, cancellation) as an auditable, timestamped record (§14.2, §21.5).

ID	Requirement
REQ-SUB-019	The system shall allow a venue owner to view their current plan, billing cycle, next renewal/expiry date, and Founding Venue/Promo status from within their account settings.
REQ-SUB-020	Social Media Manager accounts shall not have access to subscription/billing screens or actions (consistent with the existing restricted-role model, REQ-ACC-033).
REQ-SUB-021	<i>(New, v1.7)</i> The system shall grant the no-card 14-day free trial (REQ-SUB-001) at most once per Google Place ID, ever — a Place ID that has already consumed its trial shall not receive a second one on any subsequent claim, including a claim occurring after the cooldown period in REQ-ONB-024 has elapsed; a subsequent claim shall place the venue directly on the Free plan (§10.16).

8. Acceptance Criteria

Written as Given/When/Then. These are the scenarios a QA pass and Claude’s own implementation checks should run against before a feature is considered done. §8.12 restates the top-level original-scope MVP “done” bar; §8.13–§8.14 add equivalent gates for the two v1.4 capability areas.

8.1 Account creation

- *Given* a new visitor selects “Business,” *when* they attempt to sign up with a gmail.com address, *then* the system rejects the submission and explains a business email is required.
- *Given* a new visitor completes signup, *when* they have not yet clicked the emailed verification link, *then* onboarding is blocked at the verification step with a resend option.
- *Given* a returning user checked “remain signed in,” *when* 31 days have elapsed since last authentication, *then* they are required to log in again.

8.2 Venue claiming

- *Given* a venue searches Google Places for its listing, *when* the Place ID is already claimed by another Nearby Vibes account, *then* the system blocks the claim and displays “This venue is already registered.”
- *Given* a venue account has already claimed one venue, *when* the same account attempts to claim a second venue, *then* the system rejects the attempt per REQ-ONB-016.
- *Given* a venue confirms the correct Google Places match, *when* the claim completes, *then* name, address, hours, and category are pre-populated and editable only via profile completion fields (features, logo, images), and a 14-day trial is started per REQ-SUB-001.

8.3 Post lifecycle

- *Given* a venue has an Active post, *when* they attempt to publish a second post, *then* the system blocks publish and explains only one Active post is allowed per venue.
- *Given* a Scheduled post reaches its start time, *when* the system’s scheduler runs, *then* the post transitions to Active and becomes publicly visible within 60 seconds.
- *Given* an Active post reaches its end time, *when* the system’s scheduler runs, *then* the post transitions to Archived and is removed from public tile/map display within 60 seconds.
- *Given* an Archived post, *when* a venue attempts to delete it, *then* the system blocks deletion; duplication remains available (an Admin may still delete it directly, REQ-ADMIN-016).
- *Given* a Drink Special is being created, *when* the venue reaches the preview step, *then* the compliance disclaimer is visibly appended and cannot be edited or removed.

8.4 Age gating

- *Given a first-time visitor in the US views the Tile View, when drink specials are present, then their tile images render blurred with an age-gate control overlaid, while food specials render normally alongside them.*
- *Given a user confirms they are 21+ (US) or 18+ (non-US), when the confirmation is accepted, then drink specials render unblurred for the remainder of the session per the persistence rule in §10.1.*
- *Given a user declines or does not confirm age, when they browse, then drink specials remain blurred/inaccessible but food specials remain fully browsable.*

8.5 Discovery

- *Given a user has denied location permission, when they load Tile or Map View, then the system returns fuzzy, neighborhood-level results rather than an empty screen, and displays the location-benefit prompt.*
- *Given a user is within 0.75 mi of three venues with active specials, when they open Map View, then all three render with the correct category marker (drink/food) and the live-count banner reflects “3.”*

8.6 Follow lifecycle

- *Given a first-time user scans the universal QR code and it resolves to an Exact-match venue, when they view that venue’s special, then they are entered into a 48-hour trial follow and shown a confirmation prompt.*
- *Given a user in an active trial follow accumulates engagement points (§10.6) that cross the conversion threshold, when the next 2-hour scoring cycle runs, then the trial converts to a full/hard follow.*
- *Given a user’s trial follow reaches 48 hours without crossing the threshold, when the expiry job runs, then the trial follow is removed and no further pushes are sent for that venue.*
- *Given a user is already a full follower of a venue, when an Exact-match QR scan resolves to that same venue again, then the system displays “You are already following this venue” rather than re-prompting a follow.*

8.7 Push notifications

- *Given a venue publishes a new Active post, when eligible nearby/following users are computed, then a push is queued subject to the bundling window, quiet hours, and frequency caps in §15.*
- *Given a device has already received 2 pushes/bundles in the trailing 24 hours, when a new qualifying event occurs, then no further push is sent to that device until the 24-hour window resets.*
- *Given the local venue time is 1:00 AM, when a qualifying push event occurs, then the push is queued but held until 8:00 AM local time (quiet hours).*
- *Given a post expires, when 10 minutes have elapsed and the venue has not published a new Active post within 9 minutes of expiry, then a “your special has ended” push is sent to the venue owner/social manager.*

8.8 QR + arrival confirmation

- *Given a user scans the universal QR code without location sharing enabled, when the scan resolves, then the system prompts for location access before revealing any venue-specific content.*
- *Given a user scans the QR code within 30m of exactly one venue that has no live special, when the scan resolves, then the system shows “You’re at {venue}” followed by “No venue live specials right now — check back soon” and returns the user to the Home screen.*
- *Given a user scans the QR code between 30–100m of the nearest venue, when the scan resolves, then the system shows “You’re near Nearby Vibes. Find specials near you.” with a list of nearby participating venues, not a single venue’s post.*
- *Given a user scans the QR code more than 100m from any venue, when the scan resolves, then the system routes to the standard Home Tile/Map experience with no venue-specific messaging, and the scan is still logged.*

- *Given* a scan point falls within the exact-match tie-break margin of two or more venues, *when* the scan resolves, *then* the system presents a “Which venue are you at?” disambiguation list rather than guessing.
- *Given* a user viewed a post and, within 4 hours, opens the PWA while within 30m of that venue’s property-line geofence, *when* the location check runs, *then* `arrival_recorded = true` is written and reflected in venue metrics within one refresh cycle.

8.9 Navigation tracking

- *Given* a user taps “Navigate” from a post, *when* the native maps app opens, *then* a `navigation_click` event is recorded and attributed to that post/venue before the hand-off completes.

8.10 Reporting

- *Given* a user taps “Report” on a post or venue profile, *when* the report is submitted, *then* the Admin receives a push notification and a new entry appears in the report list, with a direct link and view/edit/delete actions on the reported content (REQ-ADMIN-018–019), within the same session.

8.11 Dashboards

- *Given* a venue opens their dashboard, *when* the page loads, *then* “Customers Brought In” renders as the most prominent metric, above the five supporting visuals (§17.3), *and* the dashboard is accessible only while the venue’s subscription state is Trial or Premium (§17.5, REQ-SUB-007).
- *Given* an admin opens the platform dashboard, *when* the page loads, *then* network totals (venues posting, installs, notification opt-in rate, location opt-in rate, QR performance, and subscription health) are visible without further navigation.

8.12 MVP “done” gate — original discovery/QR scope (from source concept brief, restated as acceptance criteria)

The original-scope MVP shall be considered functionally complete when all of the following hold simultaneously:

1. A user can see nearby drink and food specials, respecting the age-gate rule.
2. A user can receive a push notification for a qualifying nearby or followed-venue post.
3. A user can navigate to a venue via native maps hand-off.
4. Arrival is recorded when a user opens the PWA within 30m of a venue within 4 hours of viewing that venue’s post.
5. A QR scan (with geofence validation) results in the correct follow state (trial or “already following”) and, where thresholds are met, eligible push notifications.
6. A venue can see the full attribution funnel for a post (impressions → opens → navigation clicks → confirmed arrivals) on their dashboard.

8.13 MVP “done” gate — Subscription & Billing (new, v1.4)

The Subscription & Billing capability shall be considered functionally complete when all of the following hold simultaneously:

1. A newly claimed venue is automatically enrolled in a 14-day trial with no credit card required, and Premium features are visible/usable throughout the trial.
2. A venue that takes no action is automatically downgraded to the Free plan at hour 336 (14 days) of the trial, and Premium-only screens correctly reflect the downgrade.

3. A venue can reach pricing from the footer at any time and complete a Monthly or Annual Premium checkout, inclusive of the calculated platform service fee, via Paddle.
4. A qualifying Manhattan venue can redeem the Founding Venue Program or a Promo Code and see the 50%-off pricing reflected correctly at checkout, and the program stops being offered once the 50-venue cap is reached.
5. A successful Paddle webhook activates the subscription within the same request cycle; a failed payment leaves the venue's prior plan state untouched and surfaces a retry-capable error.
6. The venue owner receives the correct push/email at each lifecycle milestone in §15.9, and Social Media Manager accounts cannot see or act on any billing screen.

8.14 MVP “done” gate — Expanded Admin Panel (*new, v1.4*)

The Admin Panel capability shall be considered functionally complete when all of the following hold simultaneously:

1. Admin login is only reachable at the non-public portal path, requires business email + password + a valid TOTP code, and locks out after 5 consecutive failed attempts.
2. An Admin can view, edit, and delete any User account, any Venue account (including its posts and subscription state), and any individual Post.
3. A push notification and onboarding-queue entry are generated when a new Venue completes onboarding with an address resolving inside New York City.
4. A report against a Venue or a Post generates a push notification and a report-list entry with a working direct link and functioning view/edit/delete actions, without additional navigation.
5. Every admin action taken under points 2–4 is present in the admin audit log with actor, timestamp, target, and action type.

9. UI Specifications

UI specifications describe required screens, states, and components at a level sufficient to implement without the Figma file. Where the Figma reference (Nearby Vibes Claude, node 0-1) conflicts with this document, this document is authoritative. Detailed visual/look-and-feel treatment for every screen below — including the two screen groups new in v1.4 — is maintained separately in the companion **Nearby Vibes Design Brief**, which this PRD's structural requirements take precedence over if the two ever conflict.

PM Decision: the v1.0/v1.1 drafts of this document intentionally left visual style direction (color, type, iconography) open. That gap was closed in v1.3 — see §9.12 Visual Design Direction for the confirmed brand palette and reference sites. Structural specifications below (screens, states, components) remain the authoritative source for behavior; §9.12 governs look, and the Design Brief governs its execution.

9.1 Screen inventory

Screen	Primary persona	Key states
Home — Tile View	User	Default (results), fuzzy-location, empty-radius, age-gated tiles present
Home — Map View	User	Default, clustered, zoomed-out, marker-selected (preview → full)
Onboarding (permissions)	User	Notification prompt, location prompt, both-granted, both-denied
Post detail (full)	User	Image, video (paused/playing), age-gated (locked), expired-redirect
Venue profile	User	Standard, no-active-special
Account type selector	User/Venue	Personal vs. Business
Signup / Login	User/Venue	Social, email, error states (bad domain, weak password, unverified)
Venue claim search	Venue	Search, map-confirm, already-claimed error, no-results
Venue profile setup	Venue	Incomplete (blocks first post), complete
Post creation wizard	Venue/Social Manager	Type select → fields → media upload/crop → disclaimer (drink) → preview → schedule/publish
Post management list	Venue/Social Manager	Draft, Scheduled, Active, Archived filters

Screen	Primary persona	Key states
Team seat invite (<i>role type + title, v1.5; Premium-gated, v1.6</i>)	Venue	No-seat/invite-available, role-type + title selection, pending invite, accepted, revoked, seat-filled (invite blocked), Free-plan locked (upgrade required), seat suspended (venue downgraded to Free)
Venue dashboard	Venue/Social Manager	Standard (5 visuals + headline metric), Social-Manager-scoped (venue overall performance + own post detail, <i>expanded v1.5</i>)
Subscription / Pricing panel (<i>new</i>)	Venue	Trial-active countdown, Founding Venue eligible, Founding Venue cap-reached, Monthly/Annual comparison, promo-code entry/applied
Checkout (<i>new</i>)	Venue	Plan + platform-fee summary, Paddle processing, success, declined/error
QR landing	User	Location-prompt-first, exact-match (special-live / no-live-special), nearby-fallback list, out-of-range, multi-venue disambiguation
Admin console	Admin	Report queue, venue fraud queue, network dashboard
Admin portal login (<i>new</i>)	Admin	Credential entry, TOTP entry, locked-out
Admin account management (<i>new</i>)	Admin	Users list/edit/delete, Venues list/edit/delete, Posts list/edit/delete
Admin onboarding queue (<i>new</i>)	Admin	New NYC-venue alert list
Static pages	All	About / For Users / For Businesses / Terms / Privacy / AUP / Contact
PWA install prompt	User	Bottom banner, device-specific instructions
Cookie consent banner	All	First-visit, manage-preferences expanded

9.2 Home — Tile View

- Layout: single-column, vertically scrolling card list on mobile; responsive multi-column grid ≥ 768 px viewport width.
- Each tile is a single hero-image container (no grid/stacked sub-sections inside a tile, consistent with REQ-POST-014) showing, top to bottom or overlaid: hero image, promo/headline text, price or % off, venue name + address, distance badge, and a live countdown timer that visually decays (e.g., a shrinking bar or a color shift as time runs low).
- Drink-special tiles render with a blurred hero image and an age-gate badge/overlay until the user has confirmed age in-session (§10.1); food-special tiles render normally alongside them in the same list — no separate “drink” tab.
- Toggle control (Tile/Map) persists in a fixed header position across both views.
- Infinite scroll / lazy-loaded tiles; images load only as they approach the viewport (REQ-MEDIA / performance

§19).

9.3 Home — Map View

- Full-viewport map with a floating header banner: “X live specials near you.”
- Marker states per REQ-UX-014; tap → compact preview card (image, title, venue, address, time remaining) anchored above the marker; second tap → full post detail screen.
- Layer toggle control (street/satellite) and a centered “recenter to me” control.
- Clustering activates automatically once marker density crosses a defined pixel-proximity threshold at the current zoom level.

9.4 Onboarding (first-run)

- Sequential, dismissible prompts: (1) notification permission rationale + native prompt, (2) location permission rationale + native prompt. Each screen explains why before invoking the native OS permission dialog (to avoid a “just no” reflex denial).
- If both are denied, the user proceeds directly to Tile View in its fuzzy-location, no-push state rather than being blocked.

9.5 Age gate

- Presented as a lightweight in-context overlay on the blurred drink tile/post, not a full-screen interstitial that blocks the whole app.
- Copy: jurisdiction-aware (“Confirm you are 21+ (US) / 18+ (outside US)”), with Confirm and Not Now actions.
- On confirm, the age-gate state persists for the session. (*P1: persist beyond session — MVP minimum is session-level.*)

9.6 Post creation wizard (Venue / Social Manager)

- Step 1 — Type: Drink or Food.
- Step 2 — Core fields: title, hook, price/% off, description, start/end date-time, optional conditions, optional compliance/food-field note.
- Step 3 — Media: hero image (required, with crop tool), optional secondary media (image or ≤15s video, with crop/trim-to-limit tooling).
- Step 4 — Compliance (Drink only): disclaimer shown read-only, plus links to the Alcohol Advertising Help and “Tips to Boost Results” reference pages.
- Step 5 — Preview: fully rendered post exactly as users will see it, on both tile and full-post layouts.
- Step 6 — Schedule/Publish: choose Draft (save only), Scheduled (future start), or Publish Now.
- The wizard shall block advancement past a step when its required fields are incomplete, and shall preserve entered data if the venue navigates back.

9.7 Venue dashboard

- Header metric: “Customers Brought In” (current period, e.g., this week), rendered larger/bolder than any other element on the page (§17.3). Accessible only to venues in Trial or Premium subscription state (REQ-SUB-007).
- Exactly five supporting visuals, per the source brief’s explicit cap (§17.3).
- **Delegated team member view (expanded, v1.5):** renders the venue’s overall performance view (headline metric + the four network-level charts) **plus** the Post Performance Card scoped to posts they personally created (REQ-ACC-034); other creators’ individual post detail, and all billing/subscription screens, are not shown to that role (REQ-ACC-033, REQ-SUB-020). This entire view is itself a Premium-tier capability (*new, v1.6*): it is available

whenever the venue is in Trial or on paid Premium, and access suspends automatically (without deleting the underlying seat) if the venue's subscription lapses to Free (REQ-ACC-039–040, §10.15).

9.8 Subscription / Pricing panel (*new, v1.4*)

- Accessible from the global footer (“Pricing,” REQ-UX-041) at any time, and surfaced contextually during the trial per the milestone schedule in §15.9.
- Displays: current plan/trial-countdown status; a Monthly (\$39/mo) vs. Annual (\$390/yr) comparison; the Founding Venue Program panel (§10.12) when the venue is eligible and the cap has not been reached; and a promo-code entry field.
- Selecting a paid plan routes to Checkout (§9.9). See the Design Brief for full visual treatment.

9.9 Checkout (*new, v1.4*)

- Displays a clear line-item breakdown: plan price, platform service fee (5% + \$0.50, §10.13), and total charged, before the venue confirms payment via the embedded/redirected Paddle checkout.
- Success state confirms the new plan and next renewal date; error/declined state preserves the venue's prior plan and offers a retry without re-entering plan selection.

9.10 QR landing

Per the confirmed universal-QR design (§16), this screen has no venue context until the geofence resolves one:

- If location sharing is off: full-screen prompt explaining it's required to provide accurate results, with a single primary action to enable it.
- Exact match (≤ 30 m of one venue): “You're at [Venue Name]” header, then either the full post detail screen (Active post exists) or the “No venue live specials right now — check back soon” banner routing to Home Tile View.
- Nearby fallback (30–100m): “You're near Nearby Vibes. Find specials near you.” header, then a filtered list of nearby participating venues (not a single venue's post).
- Out of range (> 100 m): no special header copy; hands off directly into the standard Home Tile/Map experience.
- Multi-venue overlap at the scan point: a brief “Which venue are you at?” disambiguation list in place of the single-venue header.

9.11 Admin portal login (*new, v1.4*)

- Reached only via the non-public path defined in REQ-ADMIN-010/§20.2; not linked from any consumer-facing screen.
- Two-step form: business email + password, then a TOTP code entry step. A locked-out state displays the cool-down remaining (REQ-ADMIN-013) without revealing whether the lockout was due to a bad password or a bad TOTP code (avoids leaking which factor failed).

9.12 Admin account management & onboarding queue (*new, v1.4*)

- Three list-detail views (Users, Venues, Posts) each with search/filter, and view/edit/delete actions consistent with REQ-ADMIN-014–016.
- A dedicated Onboarding Queue view lists newly completed NYC venues (REQ-ADMIN-017) chronologically, with a link into that venue's record.
- The existing report queue (REQ-ADMIN-018–019) gains inline view/edit/delete actions directly on each list row.

9.13 Footer (global)

- Copyright line (current year, dynamically generated), links to Terms, Privacy, and Acceptable Use, a location-access toggle, a manual “Install App” trigger that re-opens the device-specific install instructions on demand, and — *new in v1.4* — a **Pricing** link routing to the Subscription/Pricing panel (§9.8).

9.14 Cookie consent banner (global)

- Bottom-anchored banner on first visit: Accept, Reject, Manage Preferences (expandable to essential/functional/analytics categories). Essential cookies are always active and not togglable.

9.15 Accessibility requirements (applies to all screens)

- WCAG 2.1 AA color contrast, focus states, and keyboard navigability across all interactive elements, including map controls, the post creation wizard, and — *new in v1.4* — the subscription/checkout flow and the admin portal.
- All non-decorative images (hero images, icons conveying state) require alt text; venue-uploaded hero images shall prompt for an optional alt-text field at upload.
- Countdown timers and other auto-updating content shall be exposed to assistive technology via appropriate ARIA live regions without being disruptively verbose.

9.16 Visual design direction

Status: Confirmed by Product. The new design shall move away from the dated reference Figma file toward something more visual and image-forward, with cleaner lines — closer to a contemporary consumer app.

Reference sites (direction, not literal templates to copy):

Reference	What to take from it
on.com zillow.com	Clean lines, minimalist layout, generous whitespace How to present a large volume of location-based cards/listings without feeling cluttered — directly relevant to Tile View (§9.2) and Map View (§9.3) at scale
alltrails.com	Overall app layout patterns — map-plus-list navigation, card structure
nps.gov digital apps	Visual tone — the illustrative, high-contrast, confidently graphic style

Brand color palette (confirmed hex values):

Swatch	Hex	Suggested role (for the designer to finalize)
Bright mint/teal	#25EFB8	Primary accent — active-special highlights, primary CTAs
Vibrant magenta	#F814E8	Secondary accent — drink-special category color, high-energy moments

Swatch	Hex	Suggested role (for the designer to finalize)
Violet/purple	#7F53F3	Tertiary accent — food-special category color, secondary interactive states
Black	#000000	Primary text, high-contrast surfaces
White	#FFFFFF	Base background

PM Decision: the role mapping above is a suggested starting point, not a locked spec — the visual design pass owns finalizing it. WCAG 2.1 AA contrast (REQ-UX-042) is still binding; #25EFB8, #F814E8, and #7F53F3 should be treated as accent/background/state colors rather than body-text color. The Admin portal (§9.11–§9.12), being an internal operations tool rather than a consumer surface, may use a plainer, higher-density visual treatment than the consumer-facing brand system — see the Design Brief for the recommended split.

10. Business Rules

10.1 Age gating (alcohol content)

- Jurisdiction is inferred from the user’s IP/locale at first gate encounter: United States → 21+; all other jurisdictions → 18+.
- Verification is self-attestation only in MVP — no ID upload or third-party age-verification service.
- Food specials shall never be gated. Drink specials shall render with a blurred hero image plus an in-context age-gate control, alongside ungated food specials in the same tile feed — never a full-screen blocking interstitial for the whole app.
- Once confirmed, the gate shall not re-prompt for the remainder of the session. Device-level persistence across sessions is a P1 enhancement, not an MVP blocker.
- A user who declines shall retain full access to food specials; only drink-special detail content remains blurred/inaccessible.

10.2 Mandatory drink-special disclaimer (verbatim, immutable)

Alcohol Promotions – Please drink responsibly In the United States, guests must be 21 years of age or older with valid government-issued identification, or meet the legal drinking age requirements applicable in their jurisdiction. NY Alcoholism assistance hotline: 877-8-HOPENY or text HOPENY to 46769, or check the health authority in your jurisdiction. See Terms of Service for more.

This text shall be appended automatically to every Drink Special post at publish time and shall not be editable or removable by the venue or Social Media Manager.

10.3 Post rules

Rule	Value
Max Active posts per venue	1 at any given time
Max post duration	7 days from first activation to final archival
Recurring daily windows	Supported within the 7-day ceiling; post is publicly visible only during defined window(s)
Push on multi-day/recurring post	Fired once, at first activation only
Archived posts	Not deletable by the venue; duplicable with a new post ID; deletable by Admin (REQ-ADMIN-016)
Draft posts	Deletable and duplicable
Media per post	1 hero image (required) + 1 optional secondary asset (1 image OR 1 video ≤15s)

10.4 Business email domain blocklist

Business accounts shall be blocked from registering with any of the following free/personal email domains (configurable list, seeded with): gmail.com, yahoo.com, hotmail.com, outlook.com, aol.com, icloud.com, live.com, msn.com, protonmail.com, mail.com

PM Decision: the source brief specified “block the top 10 common free email domains” without naming them. The list above is the seeded default; it shall be stored as configuration, not hardcoded.

10.5 Venue fraud prevention rules

Reject venue account/claim creation when any of the following hold:

1. The requesting account has already claimed a venue (one venue per business account, MVP).
2. The target Google Place ID is already claimed by another Nearby Vibes venue account.
3. The claim pattern matches automated/scripted behavior as flagged by Cloudflare Bot Management.
4. **New, v1.7:** the target Google Place ID is within its post-deletion re-claim cooldown (§10.16).

Any rejected-but-suspicious attempt shall be logged and surfaced to the Admin fraud queue rather than only silently blocked. §10.16 defines a related but distinct fraud vector — trial abuse via delete-and-re-claim — that is handled with its own persistent, account-independent ledger rather than the account-scoped checks above.

10.6 Engagement scoring model, and its use in follow conversion

Status: Confirmed by Product. This single scoring model converts a Trial follow to a Hard follow, and feeds the “highest engagement”/trending-venue ranking on the Admin dashboard (§17.4).

Qualifying action	Points
Push Open	1
Post View	1
Post Expand (tile → full post)	2
Directions Click (maps hand-off)	3
Return Session (repeat open within the window)	3
Attendance (geofenced arrival confirmation, §10.8)	3

Evaluation cadence: scores are recomputed on a 2-hour batch cycle. **Conversion threshold — confirmed:** a Trial follow converts to a Hard follow once a device/venue pair accumulates 5 or more points within the 48-hour trial window. If the threshold is not met by hour 48, the trial expires automatically.

10.7 Notification governance

- Max 2 pushes or bundles per device per rolling 24 hours.
- Max 8 pushes per device per rolling 7 days (the tighter of the two caps always wins).
- Max 2 posts per venue per day count toward push eligibility.
- Minimum 60 minutes between pushes to the same device.
- Quiet hours: no pushes between 12:00 AM–8:00 AM, venue local time.
- Notification bundling window: 15 minutes.
- Subscription-lifecycle operational pushes (§15.9) do not count against these device caps, consistent with the existing treatment of venue-owner operational pushes (§15.8).

10.8 Geofencing & arrival confirmation

- Venue geofence = property line polygon plus a 30-meter buffer, sourced/approximated at onboarding.
- Arrival confirmation fires when: the user is within the geofence, and the PWA is opened within 4 hours of that user having viewed a post for that venue.
- The 48-hour trial-follow “actionable” boundary (city boundary or Google Sublocality Level 1) is distinct from the venue-arrival geofence and governs notification eligibility, not arrival confirmation (§15.6).

10.9 Content moderation

- Any user can report a live post; any user can report a venue profile.
- Reports route to the Admin report queue with a direct reference to the content and requester context, and now include inline view/edit/delete actions (REQ-ADMIN-018–019).
- Admin has unilateral takedown authority; takedown action is logged (REQ-ADMIN-004, REQ-ADMIN-020).

10.10 Data retention & privacy baseline

- Location data collected for geofence/arrival checks is used transactionally and is not retained as a continuous location trail.
- Cookie/consent choices are stored per REQ-UX-040 and are re-solicited if the consent framework version changes.
- Static legal pages (Terms, Privacy, AUP) are the authoritative source for user-facing retention commitments.

10.11 Subscription plans & trial (*new, v1.4*)

Plan	Price	Notes
Trial	Free, 14 days	Starts automatically on venue claim (REQ-SUB-001); no credit card required; full Premium feature access (REQ-SUB-002)
Free	\$0	Default plan after trial expiry absent an upgrade; analytics dashboard and related Premium features (§17) are not accessible
Premium — Monthly	\$39/month	Full analytics access (REQ-SUB-007)
Premium — Annual	\$390/year	Equivalent to 10 months’ price for 12 months of service (~17% discount vs. 12× monthly); full analytics access

PM Decision: the source brief did not specify the exact trigger instant for “trial starts.” This document sets the trial-start trigger as **the moment the venue claim completes** (REQ-ONB-014 → Claimed state), not profile completion, since the source brief’s own wording (“Once an account is created for a venue the trial period is 14 days”) ties it to account/venue creation rather than to finishing the profile. Engineering should treat `venues.created_at` (equivalently, the Claimed-state transition, §11.3) as the trial anchor.

At trial end, absent a completed paid checkout, the system shall downgrade the venue to the Free plan (REQ-SUB-003) and shall suppress Premium-only screens/data (§17.5) starting from that transition.

10.12 Founding Venue Program & Promo Codes (*new, v1.4*)

Founding Venue Program. Displayed on the subscription/pricing panel using the confirmed program copy:

Founding Venue Program Limited to the first 50 Manhattan venues. Receive:

- 50% off for your first year
 - Founding Venue badge
 - Priority support
 - Early access to new features
 - Opportunity to influence the product roadmap
- Eligibility: venue’s claimed address resolves within Manhattan, NY (configurable to “qualifying NYC venues” more broadly, per source-brief wording — see PM Decision below).
 - Cap: the first 50 qualifying venues to redeem (REQ-SUB-009–010); the offer shall stop being displayed/offered once the cap is reached, and the count shall be tracked in real time against `promo_codes.max_redemptions` (§14.2).

Promo Code. A configurable, general-purpose promo mechanism grants 50% off either (a) the first 12 months of the Annual Premium plan, or (b) 12 consecutive monthly Premium billing cycles, after which standard pricing applies (REQ-SUB-008).

PM Decision: the source brief used both “qualifying NYC venues” (in the Promo Code section) and “first 50 Manhattan venues” (in the Founding Venue Program panel copy) without reconciling the two. This document treats the **Founding Venue Program** as the Manhattan-specific, capped-at-50 offer with the full non-monetary benefits (badge, priority support, etc.) shown above, and the **Promo Code** as the more general 50%-off discount mechanism usable for broader NYC-borough qualification or admin-issued codes, without the Founding Venue non-monetary benefits or the 50-venue cap. Both share the same 50%-off-first-12-months discount shape and the same underlying `promo_codes` table (§14.2) but are distinguished by a `program_type` flag. Product should confirm this split before launch if the intent was for these to be the same single offer.

PM Decision: the source brief’s “X12 month to month subscriptions” is interpreted as 12 consecutive monthly Premium billing cycles at the discounted rate, reverting to standard \$39/month pricing on the 13th cycle. If a different interpretation was intended, only the `promo_codes.duration_months` configuration value needs to change — no schema change is required.

- Founding Venue status, once granted, shall be visible to the venue owner in account settings (REQ-SUB-019) at minimum; whether the badge is also shown publicly on the venue’s profile is a P1 design decision, not a functional blocker (see Design Brief for the recommended treatment).

10.13 Checkout & platform service fee (*new, v1.4*)

- On checkout, the platform service fee is computed as: $\text{fee} = (\text{plan_price} \times 0.05) + 0.50$, and the total charge submitted to Paddle is $\text{plan_price} + \text{fee}$.
- Example: Monthly Premium at \$39.00 → $\text{fee} = \$39.00 \times 0.05 + \$0.50 = \$2.45$ → total charged = \$41.45.
- Example: Annual Premium at \$390.00 → $\text{fee} = \$390.00 \times 0.05 + \$0.50 = \$20.00$ → total charged = \$410.00.
- The fee is disclosed as a distinct line item at checkout (\$9.9) before the venue confirms payment — it shall never be silently bundled into an opaque “total” without a visible breakdown.
- Promo/Founding Venue discounts are applied to `plan_price` *before* the service fee is computed (i.e., the venue is never charged a service fee calculated against the undiscounted price).
- On a successful Paddle webhook, the system shall activate the subscription immediately (REQ-SUB-014); on failure, the prior plan state is retained and the venue sees a retryable error (REQ-SUB-015).

10.14 Admin portal security (*new, v1.4*)

- The Admin portal is reachable only via a distinct, non-public URL path that is never linked from consumer-facing navigation, the sitemap, or crawlable markup, and is disallowed in robots.txt (REQ-ADMIN-010, REQ-SEO-005).
- Login requires business email + password (policy per REQ-ACC-006) + a mandatory TOTP second factor (REQ-ADMIN-011).
- Cloudflare Turnstile, Bot Management, and a stricter-than-standard rate limit apply to the portal's login and MFA endpoints (REQ-ADMIN-012, §18.3).
- 5 consecutive failed attempts (password or TOTP) lock the account for a configurable cool-down and alert on-call staff (REQ-ADMIN-013).

PM Decision: the exact obscure URL path/subdomain for the Admin portal is intentionally **not published in this document** — publishing it here would defeat its purpose as a non-discoverable entry point. Engineering should provision it as a secret configuration value (environment variable, not a route constant checked into a public repository history) and share it with Admin staff out-of-band. See §20.2 for the domain-structure placeholder.

10.15 Delegated venue seat (Venue Team Member / Social Content Creator) (*new, v1.5*)

- Venue social media is typically run day-to-day by someone other than the account holder — a junior team member or an outside content creator — rather than the owner personally. The delegated-seat mechanism (§5.3, REQ-ACC-030–038) exists to give that person their own login without exposing the owner's account settings or billing.
- **Cap: exactly one delegated seat per venue account.** A venue with a pending or active invite shall not be able to send a second one; the invite screen shall communicate that the seat is filled rather than presenting a disabled control with no explanation (§9.6/Design Brief).
- **Role type is chosen by the owner before the invite is sent** — Venue Team Member or Social Content Creator — and is confirmed as a label, not a distinct permission set (Product confirmation, v1.6, Appendix A item 8): both share one identical privilege set, listed below.
- **Seat title** is a free-text field the owner sets (e.g., “Marketing Coordinator,” “Weekend Creator”) and is cosmetic — it has no effect on permissions and is not validated against a fixed list.
- **Privileges (identical for both role types, confirmed):**
 - The invited person creates their own account, using their own details (name, email, credentials) — they are not a shared login (REQ-ACC-031).
 - They can create a post for the venue, and can edit or delete only the posts they personally created — not posts created by the venue owner or by a prior seat-holder.
 - They can view the venue's overall performance dashboard (headline metric + the four network-level visuals, §17.3) in addition to the detailed Post Performance Card for posts they personally created; they cannot see other individual creators' post-level detail (i.e., a former seat-holder's post detail, if posts persist after a seat change).
 - They cannot access venue settings, follower data, subscription/billing screens, or account deletion/edit functions (REQ-ACC-033, REQ-SUB-020).
- **Seat turnover:** if the owner revokes the current seat-holder, the seat becomes available again and the owner may invite a replacement (REQ-ACC-038). Posts already created by a revoked seat-holder remain attributed to them and are not reassigned or deleted as part of revocation.
- **Premium-plan gating (*new, v1.6*):** the delegated seat is a Premium-tier capability, not a Free-plan one. It is available while the venue is in its 14-day Trial (which carries full Premium access per REQ-SUB-002) or on an active paid Premium subscription (Monthly or Annual), but **not** while the venue is on the Free plan (REQ-ACC-039).
 - A venue on the Free plan shall not be able to send a new delegated-seat invite; the invite screen shall explain that this requires Trial or Premium and shall link to the Subscription/Pricing panel (§9.8, Design Brief).
 - If a venue **already has** a Pending or Active delegated seat and its subscription lapses to Free (trial expiry

- without upgrade, REQ-SUB-003; or non-renewal, REQ-SUB-017), the system shall **suspend** — not delete — that seat’s access (REQ-ACC-040): the seat-holder’s login stops being able to create/edit/delete posts or view the dashboard for as long as the venue remains on Free.
- If the venue subsequently returns to Premium (upgrade or renewal, §11.5), the suspended seat’s access shall be **restored automatically**, with the same role type, title, and post-authorship history intact — the owner shall not need to send a new invite.
 - This gating is enforced as a runtime authorization check against the venue’s current subscription state (§21.2) at the point of each seat-scoped request, rather than as a stored “suspended” status on the seat record itself — so the seat’s `venue_staff.status` (§14.2) continues to reflect only pending/active/revoked (owner-driven actions), while Premium-gating is a separate, subscription-driven check layered on top.

PM Decision (confirmed, v1.7): the source brief states privileges once, without splitting them by role type (“Venue team member OR Social Content Creator”), and caps invites at exactly one per venue account — a tighter constraint than the unlimited/unspecified invite count implied by earlier revisions of this document (§6.2 VS-04, §7.1 REQ-ACC-030 previously). Product has confirmed the one-seat cap, the shared privilege set (Appendix A, item 8), and the suspend-and-restore behavior on a Free-plan downgrade (Appendix A, item 9) — both items are now fully closed, with no further Product input required.

10.16 Trial & re-claim abuse prevention (Google Place ID) (*new, v1.7*)

Threat model: a venue owner exhausts the 14-day free trial (§10.11), deletes the venue account, and re-claims the same physical venue (same Google Place ID) under a new email address to draw a second free trial — repeating indefinitely. This section closes that path with two linked constraints.

- **One trial per Place ID, ever (REQ-SUB-021).** Trial eligibility is tracked against the Google Place ID, not the claiming account or email address. Once a Place ID has consumed its trial, no future claim against that same Place ID — by the original owner, a new email address, or a genuinely new owner — shall receive a second free trial. A venue claiming a Place ID that has already used its trial is placed directly on the Free plan and must go through paid checkout (§21.4) to reach Premium.
- **Re-claim cooldown (REQ-ONB-024).** Independent of trial eligibility, a Google Place ID whose most recent venue account was deleted may not be re-claimed by *any* account for a configurable cooldown period, **default 90 days (~3 months)**. This is a blunter, claim-level control on top of the trial-level control above — it slows down the delete-and-recreate cycle itself, not just the trial payoff at the end of it.
- **Persistent ledger (REQ-ONB-023).** Both controls depend on state that survives venue account deletion — trial-consumed status and last-deletion timestamp are stored in a dedicated `place_claim_ledger` (§14.2) keyed by `google_place_id`, not on the venues row itself, since that row may be hard-deleted.
- **Fraud queue visibility (REQ-ONB-025).** A claim rejected for being inside the cooldown window is flagged to the Admin fraud queue (§10.5) separately from the existing duplicate-active-claim rejection (REQ-ONB-015), since a cooldown rejection is a stronger fraud signal (an already-deleted claim being retried) than a simple concurrent-claim conflict.
- **Admin override (REQ-ONB-026).** An Admin can clear the cooldown for a specific Place ID — e.g., a bar is legitimately sold and the new owner needs to claim it before 90 days have passed — logged to the audit trail. The override clears the *claim* cooldown only; it does not reinstate a trial the Place ID has already consumed (that would require a separate, explicit Admin action outside this MVP’s scope, since it re-opens the exact abuse vector this section closes).

PM Decision (confirmed, v1.8): the source brief’s example cooldown (“something like... once in 3 months”) is treated as illustrative rather than a locked figure — this document sets the default to 90 days and stores it as configuration (consistent with the pattern already used for the email-domain blocklist, §10.4), so Product or Trust & Safety can tune it post-launch without a deploy. Product has confirmed the two constraints are correctly scoped as separate: the trial-eligibility rule (one trial per Place ID, ever) is **permanent** and never resets, while the 90-day cooldown governs **only** the ability to re-claim the Place ID. No further Product input is required on this item.

11. State Diagrams

11.1 Post lifecycle

stateDiagram-v2

```
[*] --> Draft : venue starts post
Draft --> Draft : edit / save
Draft --> Scheduled : required fields complete + schedule set
Draft --> [*] : delete draft
Scheduled --> Active : start time reached
Scheduled --> Draft : venue edits before start
Active --> Archived : end time reached (auto)
Active --> Archived : venue manually unpublishes
Active --> Active : venue edits (non-schedule fields)
Archived --> Draft : duplicate (new post ID)
Archived --> [*] : (not deletable by venue — Admin may delete, REQ-ADMIN-016)
```

Notes: only one post per venue may be in the Active state at a time (§10.3). Duplication always produces a brand-new Draft with a new post ID; it never mutates the Archived source.

11.2 Follow lifecycle

stateDiagram-v2

```
[*] --> NotFollowing
NotFollowing --> TrialFollow : Exact-match QR scan, OR venue-profile/post view prompts follow
TrialFollow --> HardFollow : cumulative engagement score >= 5 pts (§10.6), evaluated on 2h cycle
TrialFollow --> Expired : 48h elapsed, threshold not met
HardFollow --> HardFollow : receives all future venue pushes (subject to §15 caps)
Expired --> TrialFollow : new qualifying trial-trigger event occurs later
HardFollow --> NotFollowing : user unfollows (manual action)
```

Note: the 2-hour evaluation cadence (§10.6) means a device can cross the conversion threshold up to ~2 hours before the state actually flips to Hard follow — an accepted latency trade-off, not a bug.

11.3 Venue account / claim verification

stateDiagram-v2

```
[*] --> Unclaimed
Unclaimed --> Searching : venue begins Google Places search
Searching --> PendingConfirmation : result selected
PendingConfirmation --> Rejected_AlreadyClaimed : Place ID already claimed (active venue)
PendingConfirmation --> Rejected_CooldownActive : Place ID re-claim within cooldown (REQ-ONB-024, new v1.7)
```

```

    PendingConfirmation --> Searching : venue rejects match, searches again
    PendingConfirmation --> Claimed : venue confirms match, no active claim and no cooldown block
    Claimed --> TrialStarted : Place ID has not yet consumed a trial (REQ-SUB-021) – 14-
day trial begins (REQ-SUB-001)
    Claimed --> FreePlanDirect : Place ID has already consumed a trial (REQ-SUB-021, new v1.7) –
no trial granted
    TrialStarted --> ProfileIncomplete : profile fields outstanding
    FreePlanDirect --> ProfileIncomplete : profile fields outstanding
    ProfileIncomplete --> ProfileComplete : required profile fields submitted
    ProfileComplete --> ProfileComplete : can now publish posts; NYC onboarding alert fires if applicable (REQ-
ADMIN-017)

```

Note: Rejected_CooldownActive and the Claimed --> FreePlanDirect branch both read from the persistent place_claim_ledger (§14.2), not from the venues table itself, since that ledger is the only record that survives a prior claim's account deletion (§10.16).

11.4 Push notification lifecycle

stateDiagram-v2

```

[*] --> EventTriggered : qualifying event (new post, follow match, subscription milestone, etc.)
    EventTriggered --> Bundling : held for up to 15 min bundling window (user-
facing discovery/follow pushes only)
    Bundling --> CapCheck : bundling window closes
    CapCheck --> Suppressed : device over 24h/7d/interval cap
    CapCheck --> QuietHoursHold : within quiet hours window
    CapCheck --> Queued : caps clear
    QuietHoursHold --> Queued : 8:00 AM local time reached
    Queued --> Sent : FCM delivery
    Sent --> Opened : user taps notification
    Sent --> Unopened : no interaction

```

Operational pushes (venue-owner post lifecycle, §15.8, and subscription lifecycle, §15.9) bypass Bundling/CapCheck — they go directly from EventTriggered to Queued, since they are not subject to the user-facing device caps (§10.7).

11.5 Subscription lifecycle (*new, v1.4*)

stateDiagram-v2

```

[*] --> TrialActive : venue claim completed (REQ-SUB-001)
    TrialActive --> PremiumActive : checkout completed (Monthly/Annual/Promo) during trial
    TrialActive --> FreePlan : 14 days elapsed, no paid plan (REQ-SUB-003)
    FreePlan --> PremiumActive : checkout completed at any time via Pricing footer link (REQ-
SUB-005, REQ-SUB-011)
    PremiumActive --> PremiumActive : renews on schedule (payment succeeds)
    PremiumActive --> PastDue : renewal payment fails (Paddle webhook)
    PastDue --> PremiumActive : retry succeeds within grace window
    PastDue --> FreePlan : grace window elapses without successful payment (REQ-SUB-017)
    PremiumActive --> FreePlan : venue cancels or subscription expires without renewal

```

See §21.2 for the full subscription state model, including grace-window duration and cancellation handling.

New, v1.6: every --> FreePlan transition above additionally suspends (without deleting) any Pending or Active delegated venue seat on that venue (REQ-ACC-040, §10.15); every FreePlan --> PremiumActive transition additionally restores full access to that seat automatically. This is a derived side effect of the subscription-state transition, not a separately triggered event — no additional state is added to this diagram for it.

12. Sequence Diagrams

12.1 Venue claim & onboarding

```
sequenceDiagram
    participant V as Venue (browser)
    participant FE as Next.js Frontend
    participant BE as NestJS Backend
    participant GP as Google Places API
    participant DB as PostgreSQL/PostGIS

    V->>FE: Enter venue search query
    FE->>BE: GET /venues/search?q=...
    BE->>GP: Places Autocomplete (category filter: food/alcohol)
    GP-->>BE: Candidate list
    BE-->>FE: Candidate list (name, address, placeId)
    V->>FE: Select candidate
    FE->>BE: POST /venues/claim { placeId }
    BE->>DB: Check placeId uniqueness
    alt already claimed
        DB-->>BE: existing venue found
        BE-->>FE: 409 "already registered"
    else available
        BE->>GP: Place Details (hours, category)
        GP-->>BE: Full details
        BE->>DB: Create venue (state=Claimed, profile=Incomplete)
        BE->>DB: Create subscription (state=Trial, trialEndsAt=+14d) [REQ-SUB-001]
        DB-->>BE: venueId
        BE-->>FE: 201 { venueId, prefilled fields, trialEndsAt }
    end
    FE-->>V: Show map confirmation + profile setup form
```

12.2 Post creation & publish

```
sequenceDiagram
    participant Ven as Venue/Social Manager
    participant FE as Frontend Wizard
    participant BE as NestJS Backend
    participant MP as Media Processor (Sharp)
    participant R2 as Object Storage (R2/Spaces)
    participant DB as PostgreSQL

    Ven->>FE: Fill post fields + upload hero image
```

```

FE->>BE: POST /media/upload
BE->>MP: Validate + compress + generate thumb/feed/full
MP->>R2: Store WebP derivatives
R2-->>MP: Storage URLs
MP-->>BE: Derivative URLs
BE-->>FE: mediaAssetId + preview URLs
Ven->>FE: Confirm preview, choose Schedule/Publish Now
FE->>BE: POST /posts { fields, mediaAssetId, schedule }
BE->>DB: Validate 1-active-post-per-venue rule
alt violates active-post rule
    BE-->>FE: 409 conflict
else valid
    BE->>DB: Insert post (state=Scheduled or Active)
    DB-->>BE: postId
    BE-->>FE: 201 { postId, state }
    Note over BE: If Active immediately, triggers Push Engine (12.3)
end

```

12.3 Push notification delivery (discovery + bundling)

sequenceDiagram

```

participant Sched as Bundling Scheduler (15-min tick)
participant BE as NestJS Backend
participant DB as PostgreSQL/PostGIS
participant FCM as Firebase Cloud Messaging
participant U as User Device

Sched->>BE: Run bundling cycle
BE->>DB: Query posts activated since last cycle
BE->>DB: Match eligible devices (within 0.75mi OR hard-follow)
DB-->>BE: Candidate device list + matched posts
BE->>BE: Apply caps (24h/7d/interval/quiet-hours) per device
BE->>BE: Group multiple posts per device into one bundle
BE->>FCM: Send bundled push payload
FCM->>U: Deliver notification ("4 drink specials went live near you")
U-->>FE: Tap notification
Note over U: Opens filtered Tile View containing only bundled posts
BE->>DB: Log push_sent, push_opened events (timestamped)

```

12.4 QR scan → geofence resolution → follow/arrival

sequenceDiagram

```

participant U as User Device
participant FE as PWA
participant BE as NestJS Backend
participant DB as PostgreSQL/PostGIS

U->>FE: Scan universal QR (opens nearbyvibes.com/qr)
FE->>FE: Check location permission
alt not granted
    FE-->>U: Prompt: location required for accurate results
else granted
    FE->>BE: GET /qr/resolve { lat, lng }

```

```

BE-->>DB: Match device coords against venue geofence layer
BE-->>DB: Log qr_scan event (timestamped, resolved_venue_id, distance_tier)
alt multiple venues within exact-match tie-break margin
  BE-->>FE: disambiguation_required { candidateVenues }
  FE-->>U: "Which venue are you at?" list
else exact match (<=30m, single venue)
  alt no live special
    BE-->>FE: no_active_post { venueName }
    FE-->>U: "You're at {venue}" + "No live specials right now" banner -> Home
  else live special exists
    BE-->>FE: post payload + venueName
    FE-->>U: "You're at {venue}" + render full post
    BE-->>DB: Evaluate follow state (NotFollowing/Trial/Hard)
    alt NotFollowing
      BE-->>DB: Create TrialFollow
      FE-->>U: "You're now following this venue" prompt
    else threshold already met this session
      BE-->>DB: Convert to HardFollow ($10.6)
      FE-->>U: "You're now following this venue" prompt
    else already Trial or Hard
      FE-->>U: "You are already following this venue" prompt
    end
  end
end
else nearby fallback (30-100m)
  BE-->>FE: nearby_venues[]
  FE-->>U: "You're near Nearby Vibes. Find specials near you." + venue list
else out of range (>100m)
  BE-->>FE: out_of_range
  FE-->>U: Standard Home Tile/Map experience
end
end
end

```

12.5 Arrival confirmation on PWA open

sequenceDiagram

```

participant U as User Device
participant FE as PWA
participant BE as NestJS Backend
participant DB as PostgreSQL/PostGIS

U->>FE: Opens installed PWA
FE->>FE: Check: was a post viewed in last 4 hours?
alt no recent post view
  FE-->>U: Normal Home load, no arrival check
else recent post view exists
  FE->>BE: POST /arrivals/check { lat, lng, postId }
  BE->>DB: Distance to venue geofence (property line + 30m)
  alt distance <= 30m
    BE->>DB: Write arrival_confirmations (postId, venueId, userId, timestamp)
    BE-->>FE: arrival_recorded = true
  else outside range
    BE-->>FE: arrival_recorded = false
  end
end

```

end

12.6 Subscription checkout & Paddle payment (*new, v1.4*)

sequenceDiagram

```

    participant Ven as Venue Owner
    participant FE as Frontend (Pricing/Checkout)
    participant BE as NestJS Backend
    participant PD as Paddle.com
    participant DB as PostgreSQL

    Ven->>FE: Select plan (Monthly/Annual) or enter Promo Code, from Pricing panel
    FE->>BE: POST /subscriptions/checkout { planType, promoCode? }
    BE->>DB: Validate promo eligibility (Founding Venue cap, city qualification, code active/unused)
    BE->>BE: Apply discount (if any) to plan_price, then compute platform service fee = plan_price*0.05 + 0.50
    BE-->>FE: checkoutSessionId, planPrice, fee, totalAmount, paddleCheckoutUrl
    FE-->>Ven: Show line-item breakdown (plan + fee = total)
    Ven->>FE: Confirm, proceed to Paddle
    FE->>PD: Redirect/embed Paddle checkout with totalAmount
    Ven->>PD: Completes payment
    PD-->>BE: Webhook: payment succeeded/failed (signed payload)
    BE->>BE: Verify Paddle webhook signature
    alt payment succeeded
        BE->>DB: Activate subscription (state=Active/PremiumActive, plan, renewsAt)
        BE->>DB: Increment promo_codes.redemption_count if promo applied
        BE->>DB: Log subscription_event (upgraded / promo_applied)
        BE-->>Ven: Push + email confirmation (REQ-SUB-016)
    else payment failed
        BE->>DB: Log subscription_event (payment_failed)
        BE-->>FE: checkout_failed
        FE-->>Ven: Error + retry option (plan state unchanged, REQ-SUB-015)
    end
end

```

12.7 Admin portal login (MFA) (*new, v1.4*)

sequenceDiagram

```

    participant A as Admin
    participant FE as Admin Portal Frontend
    participant BE as NestJS Backend
    participant CF as Cloudflare (Turnstile/Bot Mgmt/Rate Limit)

    A->>FE: Navigate to non-public admin portal path
    FE->>CF: Turnstile challenge
    CF-->>FE: Challenge token
    A->>FE: Enter business email + password
    FE->>BE: POST /admin/auth/login { email, password, turnstileToken }
    BE->>CF: Verify Turnstile token + bot score
    alt too many recent failures
        BE-->>FE: 423 locked, cooldownRemaining
    else within limits
        BE->>BE: Verify email + password hash
        alt invalid credentials
            BE-->>FE: 401 (generic – does not reveal which factor failed)
        else valid credentials
            BE->>FE: 200 OK
        end
    end
end

```

```
    else valid
      BE-->>FE: mfaChallengeRequired = true
      A->>FE: Enter TOTP code from authenticator app
      FE->>BE: POST /admin/auth/mfa/verify { totpCode }
      alt valid TOTP
        BE-->>FE: accessToken (admin-scoped JWT)
      else invalid TOTP
        BE-->>FE: 401, increments failed-attempt counter
      end
    end
  end
end
```

13. API Contracts

REST/JSON over HTTPS. All authenticated endpoints require a bearer JWT (§18 Security). Paths are illustrative of resource shape and shall be finalized against the framework's routing conventions (NestJS) during implementation; field names are the binding contract.

13.1 Auth & accounts

Method & Path	Auth	Request (key fields)	Response (key fields)
POST /auth/signup	Public	accountType (personal business), method (google apple email), email?, password?	userId, verificationRequired: true
POST /auth/login	Public	method, email?, password?, idToken?	accessToken, refreshToken, expiresAt
POST /auth/logout	Bearer	—	204
POST /auth/forgot-password	Public	email	202 (always, to avoid account enumeration)
POST /auth/reset-password	Public (token)	token, newPassword	200
GET /auth/verify-email/:token	Public (token)	—	200 / redirect to onboarding continuation
GET /me	Bearer	—	userId, accountType, roles[], venueId?

13.2 Venues

Method & Path	Auth	Request (key fields)	Response (key fields)
GET /venues/search?q=	Bearer (business)	query string	[{ placeId, name, address, category }]

Method & Path	Auth	Request (key fields)	Response (key fields)
POST /venues/claim	Bearer (business)	placeId	venueId, state: Claimed, prefilled fields, trialEndsAt (omitted if the Place ID already consumed its trial, REQ-SUB-021 — venue starts on Free instead), or 409 if actively claimed, or 423 with cooldown-RemainingDays if inside the re-claim cooldown (REQ-ONB-024, new v1.7)
GET /venues/:venueId	Public	—	Full public venue profile
PATCH /venues/:venueId/profile	Bearer (owner)	logoUrl?, bgImageUrl?, features[]?	200 updated profile
POST /venues/:venueId/social-managers/invite	Bearer (owner)	email, roleType (venue_team_member social or 409 if the venue's one seat is already pending/active (REQ-ACC-035)	inviteId, status: pending, or 409 if the venue's one seat is already pending/active (REQ-ACC-035)
POST /venues/:venueId/social-managers/inviteId/revoke	Bearer (owner)	—	204, frees the seat for a new invite (REQ-ACC-038)
GET /venues/:venueId/social-managers	Bearer (owner)	—	[{ userId, email, roleType, title, status }]
POST /venues/:venueId/report	Bearer	reason?	201, notifies Admin (REQ-ADMIN-018)

13.3 Posts & media

Method & Path	Auth	Request (key fields)	Response (key fields)
POST /media/upload	Bearer (venue/social-manager)	multipart file	mediaAssetId, thumbUrl, feedUrl, fullUrl
POST /posts	Bearer (venue/social-manager)	full post fields (§7.4) + mediaAssetId	postId, state: Draft
GET /posts/:postId	Public (if Active/Archived) / Bearer (Draft/Scheduled)	—	Full post payload
PATCH /posts/:postId	Bearer (owner/social-manager)	changed fields	200 updated post
POST /posts/:postId/publish	Bearer (owner/social-manager)	scheduleAt?	200, state: Scheduled Active, 409 if active-post-limit violated
POST /posts/:postId/unpublish	Bearer (owner/social-manager)	—	200, state: Archived
POST /posts/:postId/duplicate	Bearer (owner/social-manager)	—	new postId, state: Draft

Method & Path	Auth	Request (key fields)	Response (key fields)
DELETE /posts/:postId	Bearer (owner/social-manager)	—	204 (Draft only; 409 otherwise)
GET /venues/:venueId/posts?state={owner/social-manager}	Bearer	—	[{ postId, state, title, ... }]
POST /posts/:postId/report	Bearer	reason?	201, notifies Admin (REQ-ADMIN-018)

13.4 Discovery

Method & Path	Auth	Request (key fields)	Response (key fields)
GET /discovery/tiles?lat=&lng=&(fuzzy)	Bearer or anonymous	coords, radius (default 0.75mi)	[{ postId, tileFields, distanceMeters }]
GET /discovery/map?bounds=	Bearer or anonymous (fuzzy)	bounding box	[{ postId, lat, lng, markerState }], clustered server-side

13.5 Follow

Method & Path	Auth	Request (key fields)	Response (key fields)
POST /venues/:venueId/follow	Bearer	source (qr profile post)	followState: Trial Hard, message
GET /users/me/follows	Bearer	—	[{ venueId, followState, expiresAt? }]
DELETE /venues/:venueId/follow	Bearer	—	204

13.6 QR & geofence

Method & Path	Auth	Request (key fields)	Response (key fields)
GET /qr/resolve	Bearer or anonymous (location required)	lat, lng	tier: exact nearby_fallback out_of_range, venue?, candidateVenues?, activePost?, followState?
POST /arrivals/check	Bearer	postId, lat, lng	arrivalRecorded: boolean

13.7 Devices & push

Method & Path	Auth	Request (key fields)	Response (key fields)
POST /devices/register	Bearer	fcmToken, platform	204
POST /devices/unregister	Bearer	fcmToken	204

Push dispatch itself (bundling, cap enforcement, quiet-hours evaluation) is an internal scheduled job, not a public API — see §15.

13.8 Metrics

Method & Path	Auth	Request (key fields)	Response (key fields)
GET /venues/:venueId/metrics/summary?period=	Bearer (owner)	period	customersBroughtIn, 5-visual dataset (§17.3) — 403 if venue is on the Free plan (REQ-SUB-007)
GET /venues/:venueId/metrics/posts/:postId/social-manager,	Bearer (owner or social-manager, device-scoped)	—	Post Performance Card fields
POST /events/track	Bearer or anonymous (device-scoped)	eventType, postId?, venueId?	202 (fire-and-forget event ingestion)
GET /admin/metrics/network	Bearer (admin)	—	Network dashboard dataset, including subscription health (§17.4)

13.9 Static pages & admin

Method & Path	Auth	Request (key fields)	Response (key fields)
GET /pages/:slug	Public	—	rendered page content + version
PATCH /admin/pages/:slug	Bearer (admin)	content, videoEmbeds?	new version created
GET /admin/reports	Bearer (admin)	filters	[{ reportId, targetType, targetId, targetLink, reportedAt }]
POST /admin/reports/:reportId/resolve	Bearer (admin)	action (remove dismiss edit), note?	200, logs to audit trail
GET /admin/venues/fraud-queue	Bearer (admin)	—	[{ venueId, flagReason }]

13.10 Subscriptions & billing (new, v1.4)

Method & Path	Auth	Request (key fields)	Response (key fields)
GET /subscriptions/plans	Public	—	[{ planType, price, features }]
GET /venues/:venueId/subscription	Bearer (owner)	—	{ planType, state, trialEndsAt?, currentPeriodEnd?, isFoundingVenue?, promoApplied? }

Method & Path	Auth	Request (key fields)	Response (key fields)
POST /subscriptions/checkout	Bearer (owner)	planType (monthly annual), promoCode?	{ checkoutSessionId, planPrice, fee, totalAmount, paddleCheckoutUrl }
POST /webhooks/paddle	Paddle signature (not bearer)	Paddle payload	200 (internal; verifies webhook signature before processing)
POST /venues/:venueId/subscription/cancel	Bearer (owner)	—	200, state transitions to Cancelled at period end (P1)
GET /promo- codes/:code/validate	Bearer (owner)	—	{ valid: boolean, programType, discountPercent, capRemaining? }

13.11 Admin account & content management (new, v1.4)

Method & Path	Auth	Request (key fields)	Response (key fields)
POST /admin/auth/login	Public (portal path only)	email, password, turnstileToken	mfaChallengeRequired: true, or 401/423
POST /admin/auth/mfa/verify	Public (portal session)	totpCode	accessToken (admin-scoped)
GET /admin/users	Bearer (admin)	filters	[{ userId, email, accountType, createdAt }]
PATCH /admin/users/:userId	Bearer (admin)	fields	200
DELETE /admin/users/:userId	Bearer (admin)	—	204
GET /admin/venues	Bearer (admin)	filters	[{ venueId, name, city, subscriptionState }]
PATCH /admin/venues/:venueId	Bearer (admin)	fields	200
DELETE /admin/venues/:venueId	Bearer (admin)	—	204
GET /admin/posts	Bearer (admin)	filters	[{ postId, venueId, state }]
PATCH /admin/posts/:postId	Bearer (admin)	fields	200
DELETE /admin/posts/:postId	Bearer (admin)	—	204
GET /admin/onboarding-queue	Bearer (admin)	—	[{ venueId, name, completedAt, city }] (NYC-qualified new venues, REQ-ADMIN-017)

14. Data Model

14.1 Entity relationship diagram

```
erDiagram
    USER ||--o{ FOLLOW : creates
    USER ||--o{ DEVICE : registers
    USER ||--o{ REPORT : files
    VENUE ||--o{ POST : publishes
    VENUE ||--o{ FOLLOW : "is followed via"
    VENUE ||--o{ VENUE_STAFF : employs
    USER ||--o{ VENUE_STAFF : "is invited as"
    VENUE ||--|| SUBSCRIPTION : has
    SUBSCRIPTION ||--o{ SUBSCRIPTION_EVENT : logs
    SUBSCRIPTION }o--o| PROMO_CODE : redeems
    POST ||--o{ MEDIA_ASSET : uses
    POST ||--o{ PUSH_EVENT : triggers
    POST ||--o{ QR_SCAN : "resolved via"
    POST ||--o{ ARRIVAL_CONFIRMATION : generates
    VENUE ||--o{ QR_SCAN : receives
    USER ||--o{ QR_SCAN : performs
    USER ||--o{ ARRIVAL_CONFIRMATION : triggers
    VENUE ||--o{ REPORT : "is target of"
    POST ||--o{ REPORT : "is target of"
    ADMIN_USER ||--o{ AUDIT_LOG : performs
    ADMIN_USER ||--o{ REPORT : resolves
    PLACE_CLAIM_LEDGER }o--o| VENUE : "tracks claim/trial history for"
    ADMIN_USER ||--o{ PLACE_CLAIM_LEDGER : "may override cooldown on"
```

14.2 Core tables

users

Column	Type	Notes
id	uuid PK	
account_type	enum(personal, business)	
auth_method	enum(google, apple, email)	
email	citext unique	
email_verified_at	timestampz nullable	
password_hash	text nullable	null for social-only accounts
created_at / updated_at	timestampz	

venues

Column	Type	Notes
id	uuid PK	
owner_user_id	uuid FK → users.id	one venue per owner (MVP rule)
google_place_id	text unique (among non-deleted venues)	enforces REQ-ONB-014; historical trial/claim state for this value survives row deletion in place_claim_ledger (§14.2, new v1.7), not here
name, address, category	text	imported at claim
city	text	derived from address at claim; used for REQ-ADMIN-017 NYC check and §10.12 program eligibility
opening_hours	jsonb	imported at claim
features	text[]	selected from preset list
logo_url, background_url	text	
geom	geography(Point, 4326)	PostGIS
geofence_polygon	geography(Polygon, 4326)	property line + 30m buffer, precomputed
profile_state	enum(incomplete, complete)	
created_at / updated_at	timestampz	trial anchor is created_at (§10.11)
deleted_at	timestampz nullable	(new, v1.7) soft-delete marker; on hard or soft deletion, the system shall write/update the corresponding place_claim_ledger row (below) before the venue row becomes unavailable, so trial/cooldown history is never lost

place_claim_ledger (new, v1.7 — persistent, survives venue account deletion; backs §10.16)

Column	Type	Notes
google_place_id	text PK	one row per Place ID ever claimed, regardless of how many times
trial_consumed	boolean default false	set to true the first time a trial is granted against this Place ID (REQ-SUB-001) and never reset — enforces REQ-SUB-021
trial_consumed_at	timestampz nullable	
current_venue_id	uuid FK → venues.id, nullable	null when not currently claimed
last_claimed_at	timestampz nullable	
last_deleted_at	timestampz nullable	set when current_venue_id's venue account is deleted; drives the REQ-ONB-024 cooldown window
cooldown_override_admin_id	uuid FK → admin_users.id, nullable	set when an Admin clears the cooldown (REQ-ONB-026)
cooldown_override_at	timestampz nullable	

Column	Type	Notes
created_at / updated_at	timestampz	

venue_staff (delegated team-member/content-creator seat grants)

Column	Type	Notes
id	uuid PK	
venue_id	uuid FK → venues.id	a partial unique index on (venue_id) WHERE status IN ('pending', 'active') enforces the one-seat-per-venue cap (REQ-ACC-035)
user_id	uuid FK → users.id	
role_type	enum(venue_team_member, social_content_creator)	(new, v1.5) owner's selection at invite time (REQ-ACC-036); does not change the privilege set
title	text nullable	(new, v1.5) free-text seat title set by the owner (REQ-ACC-037), e.g. "Marketing Coordinator"
status	enum(pending, active, revoked)	
invited_at, accepted_at, revoked_at	timestampz nullable	

posts

Column	Type	Notes
id	uuid PK	
venue_id	uuid FK → venues.id	
created_by_user_id	uuid FK → users.id	owner or social manager
type	enum(drink, food)	
state	enum(draft, scheduled, active, archived)	
title, hook, description	text	length-constrained per §7.4
price_or_discount	text	
conditions, compliance_note	text nullable	
start_at, end_at	timestampz	
recurrence_windows	jsonb nullable	daily time-of-day windows, if used
hero_media_id	uuid FK → media_assets.id	
secondary_media_id	uuid FK → media_assets.id, nullable	
disclaimer_applied	boolean	true for all drink posts
slug	text unique	REQ-SEO-002 pattern
duplicated_from_post_id	uuid nullable	
created_at / updated_at	timestampz	

media_assets

Column	Type	Notes
id	uuid PK	
venue_id	uuid FK	

Column	Type	Notes
type	enum(image, video)	
thumb_url, feed_url, full_url	text	WebP/MP4 derivatives
duration_seconds	int nullable	video only, ≤15
created_at	timestampz	

follows

Column	Type	Notes
id	uuid PK	
user_id	uuid FK	
venue_id	uuid FK	
state	enum(trial, hard, expired)	
engagement_points	int default 0	accumulates during trial (§10.6)
trial_started_at, trial_expires_at	timestampz	
converted_at	timestampz nullable	
created_at / updated_at	timestampz	

devices

Column	Type	Notes
id	uuid PK	
user_id	uuid FK nullable	may register before login (anonymous discovery)
fcm_token	text unique	
platform	enum(ios, android, desktop)	
notifications_enabled	boolean	
location_enabled	boolean	
pwa_installed	boolean	derived from display-mode: standalone detection
last_seen_at	timestampz	

push_events

Column	Type	Notes
id	uuid PK	
device_id	uuid FK	
bundle_id	uuid	groups co-delivered posts
post_ids	uuid[]	
sent_at	timestampz	
opened_at	timestampz nullable	
suppressed_reason	text nullable	cap/quiet-hours diagnostics

qr_scans

Column	Type	Notes
id	uuid PK	
resolved_venue_id	uuid FK nullable	null when distance_tier = out_of_range

Column	Type	Notes
distance_tier	enum(exact, nearby_fallback, out_of_range)	
candidate_venue_ids	uuid[] nullable	populated only for the multi-venue disambiguation path
user_id	uuid FK nullable	
device_id	uuid FK	
scanned_at	timestampz	

arrival_confirmations

Column	Type	Notes
id	uuid PK	
post_id	uuid FK	
venue_id	uuid FK	
user_id	uuid FK	
distance_meters	numeric	
post_viewed_at	timestampz	
confirmed_at	timestampz	

reports

Column	Type	Notes
id	uuid PK	
target_type	enum(post, venue)	
target_id	uuid	
reporter_user_id	uuid FK nullable	
status	enum(open, removed, dismissed)	
created_at, resolved_at	timestampz	

static_pages / static_page_versions

Column	Type	Notes
slug	text PK	
current_version_id	uuid FK	
version_id, content (rich text/HTML), video_embeds jsonb, edited_by, edited_at	—	version history table

admin_audit_log

Column	Type	Notes
id	uuid PK	
admin_user_id	uuid FK → admin_users.id	
action	text	e.g., report.remove, page.edit, venue.delete, user.edit
target_type, target_id	text, uuid	
reason	text nullable	

Column	Type	Notes
created_at	timestampz	

admin_users (*new, v1.4 — separate from users, backs the dedicated Admin portal, §18.1*)

Column	Type	Notes
id	uuid PK	
business_email	citext unique	
password_hash	text	policy per REQ-ACC-006
mfa_secret	text (encrypted at rest)	TOTP seed
mfa_enabled	boolean	shall be true before an admin_users row can authenticate (REQ-ADMIN-011)
failed_login_count	int default 0	
locked_until	timestampz nullable	REQ-ADMIN-013
role	enum(admin)	single role in MVP; extensible later
created_at / updated_at	timestampz	

subscriptions (*new, v1.4*)

Column	Type	Notes
id	uuid PK	
venue_id	uuid FK → venues.id, unique	one subscription record per venue
plan_type	enum(free, premium_monthly, premium_annual)	
state	enum(trial, active, past_due, expired, cancelled)	
is_founding_venue	boolean default false	
promo_code_id	uuid FK → promo_codes.id, nullable	
trial_started_at, trial_ends_at	timestampz	anchor = venues.created_at (§10.11)
current_period_start, current_period_end	timestampz nullable	null while state=trial
paddle_subscription_id	text nullable	Paddle's subscription reference, once created
created_at / updated_at	timestampz	

promo_codes (*new, v1.4*)

Column	Type	Notes
id	uuid PK	
code	text unique	
program_type	enum(founding_venue, general_promo)	distinguishes the capped Manhattan program from general NYC promo codes (§10.12)
discount_percent	int default 50	

Column	Type	Notes
duration_months	int default 12	
max_redemptions	int nullable	e.g., 50 for Founding Venue; null = uncapped
redemption_count	int default 0	
eligible_city_scope	text	e.g., “Manhattan” or “NYC”
active	boolean default true	
created_at	timestampz	

subscription_events (*new, v1.4 — append-only, mirrors the analytics event-stream pattern, §14.3*)

Column	Type	Notes
id	uuid PK	
venue_id	uuid FK	
event_type	enum(trial_started, trial_expired, upgraded, downgraded, promo_applied, renewed, payment_failed, cancelled)	
metadata	jsonb	e.g., planType, amount, paddleEventId
occurred_at	timestampz	

14.3 Analytics event stream (append-only, drives §17)

Column	Type	Notes
id	uuid PK	
event_type	enum(post_impression, post_open, venue_profile_view, navigation_click, share, qr_scan, arrival_confirmation, push_sent, push_opened)	
post_id, venue_id, user_id, device_id	uuid nullable	populated per event type
occurred_at	timestampz	
metadata	jsonb	free-form (e.g., distance, source screen)

PM Decision: the source brief lists metrics as bare field names without specifying storage shape. Modeling them as a single append-only event stream (rather than pre-aggregated counters) is the standard pattern for this kind of funnel analytics — it keeps every dashboard number re-derivable and auditable. Subscription lifecycle events use a parallel, separately namespaced stream (`subscription_events`) rather than being folded into this table, since they are billing-state transitions rather than user-behavior events and have a different retention/audit sensitivity (§18.6).

15. Notification Engine

15.1 Trigger types

Trigger	Description	Reference
General Nearby Push (discovery signal)	Fires when a special is posted within 0.75mi/1.2km of a device that is not necessarily following the venue	REQ-PUSH
Venue Follow Push (primary signal)	Fires whenever a followed venue (trial or hard) posts a new special, evaluated up to the city-boundary actionability limit	REQ-PUSH
Post-active confirmation	Sent to the venue owner/social manager when their post transitions to Active	REQ-POST-022
Post-expired reminder	Sent to the venue owner/social manager 10 minutes after a post archives, suppressed if a new Active post was created within 9 minutes of expiry	source brief, verbatim timing
Subscription lifecycle (<i>new</i>)	Trial welcome, mid-trial nudge, trial-ending warning, trial-expired/downgrade, renewal-upcoming, payment-failed, subscription-expired notices	§15.9, REQ-SUB-004, REQ-SUB-016

15.2 Eligibility computation (per trigger cycle)

1. Identify posts that became Active since the last cycle.
2. For each post, compute the eligible device set: devices within 0.75mi/1.2km of the venue (discovery signal), and/or devices with a Trial or Hard follow on the venue, evaluated against the trial's actionable boundary.
3. De-duplicate devices eligible via both signals — one bundle slot per device per cycle, not one push per trigger type.

15.3 Bundling algorithm

- Bundling window: 15 minutes. All qualifying posts for a device within the window are collapsed into a single push.

- Bundling priority order: (1) expiring specials, (2) closest venues, (3) highest-engagement venues, (4) time-sensitive events.
- Copy pattern: single post gets venue-specific copy; multiple posts get aggregate copy (e.g., “4 drink specials went live near you”).
- Tapping the notification opens a filtered Tile View containing only the posts in that specific bundle — not the general feed.

15.4 Device protection caps (enforced in this order)

1. Minimum 60-minute interval since the device’s last push.
2. Max 2 pushes/bundles within the trailing 24 hours.
3. Max 8 pushes within the trailing 7 days.
4. Max 2 qualifying posts per venue per day count toward eligibility.
5. Quiet hours: no delivery between 12:00 AM–8:00 AM, venue local time; held and delivered at 8:00 AM.

Any push blocked by caps 1–4 is dropped (not deferred); only the quiet-hours cap defers rather than drops.

15.5 Multi-day / recurring post handling

A post with recurrence windows shall trigger the discovery/follow push **once**, at first activation only. Subsequent daily windows opening shall not re-trigger a push.

15.6 48-hour trial-follow notification scope

While a follow is in Trial state, the user is eligible to receive pushes from that venue even outside the 0.75mi radius, as long as the device remains within the venue’s city boundary or Google Sublocality Level 1 region. This is the only sanctioned exception to the proximity principle in §3.

15.7 Delivery infrastructure

- Firebase Cloud Messaging (Web Push) is the delivery channel for both iOS (via installed PWA + APNs bridge through FCM) and Android.
- The PWA’s service worker shall handle push events, display the notification, and route the click-through to the filtered Tile View.
- Devices register/unregister FCM tokens via `/devices/register` / `/unregister`; token churn shall not create duplicate device rows.

15.8 Venue-owner operational pushes

Two operational pushes go to the venue owner or Social Media Manager:

- **Post went live:** “Your special has started. Remember to get people to scan your QR code.” — sent at Active transition.
- **Post expired:** “Your special has ended. Create a new offer to keep traffic flowing.” — sent 10 minutes after expiry, suppressed if a new Active post exists within 9 minutes of that expiry.

These operational pushes do not count against the user-facing device caps in §15.4 (different recipient, different purpose).

15.9 Subscription lifecycle notifications (*new, v1.4*)

Distinct from user-facing discovery/follow pushes and from the post-lifecycle operational pushes above, the following go to the **venue owner only** (never the Social Media Manager, per REQ-SUB-020) via both push and email:

Milestone	Timing	Copy intent
Trial welcome	At trial start (venue claim completion, REQ-SUB-001)	Confirms the 14-day trial and no-credit-card requirement; introduces Premium features unlocked during trial
Mid-trial nudge	Day 7 of trial	Highlights the analytics the venue has generated so far, as the reason to consider upgrading
Trial ending soon	Day 12 of trial (48 hours before expiry)	Warns that Premium access ends in 2 days; links directly to the Pricing panel
Trial expired / downgraded	Day 14, at the moment of downgrade	Confirms the move to the Free plan and that Premium analytics are no longer visible; links to Pricing
Renewal upcoming	3 days before a paid subscription's <code>current_period_end</code>	Confirms the upcoming charge amount (plan + fee) and billing cycle
Payment failed	Immediately on a Paddle webhook reporting failure	Explains the failed charge and provides a retry link; subscription enters <code>past_due</code> (§21.2)
Subscription expired / downgraded	On non-renewal after the grace window (§21.2)	Confirms the move to the Free plan

These operational pushes do not count against the user-facing device caps in §15.4/§10.7, consistent with the existing treatment of venue-owner operational pushes in §15.8, and bypass the 15-minute bundling window (§11.4) since each is tied to a specific, time-sensitive billing event rather than discovery content.

16. QR Flow

Status: Confirmed by Product. Nearby Vibes issues one universal QR code shared by every venue. The code carries no venue identifier; the venue is resolved entirely from the scanning device’s GPS position against the venue geofence layer.

16.1 Issuance

All venues share a single QR code encoding `https://nearbyvibes.com/qr`. QR codes are static (never expire, never regenerate) so venues can print them once.

16.2 Scan resolution logic

1. Device scans the code and opens `/qr`.
2. The PWA checks location-sharing status; if disabled, a blocking prompt is shown before any venue-specific content is revealed.
3. Once location is available, the backend resolves the device’s coordinates against the venue geofence layer and returns one of three distance-tiered outcomes:

Tier	Distance	Behavior
Exact match	$\leq 30\text{m}$ of a venue geofence	“You’re at [Venue Name]” — render that venue’s Active post directly, or the no-live-special banner if it has none.
Nearby fallback	30–100m	“You’re near Nearby Vibes. Find specials near you.” — render a filtered list of nearby participating venues.
Out of range	$> 100\text{m}$	No venue-specific messaging; hands off to the standard discovery experience. Scan is still logged.

4. **Overlap handling:** the system resolves to the nearest venue by distance to centroid; ties within a 5m margin trigger a “Which venue are you at?” disambiguation list.
5. A `qr_scans` row is written on every scan regardless of outcome.
6. No Active post → “No venue live specials right now — check back soon,” routed to Home Tile View. Active post exists → render the full post detail screen.
7. Follow-state branch (Exact-match only): new Trial follow, Hard-follow confirmation, or “You are already following this venue.”

16.3 Metrics captured per scan

Field	Purpose
scanned_at	Time-of-day QR-effectiveness analysis
resolved_venue_id (nullable)	Attribution — null in the Out-of-range tier
distance_tier	exact nearby_fallback out_of_range
resulting_follow_state	Trial created / Hard converted / already-following / not applicable
device_id (+ user_id if authenticated)	De-duplication and funnel joins

16.4 Edge cases

- Re-scanning while already viewing an Active post: fresh qr_scans row, no duplicate follow record.
- A scan landing in the 30–100m fallback tier does not unlock that specific venue’s special.
- QR scans do not, by themselves, satisfy arrival confirmation — that has its own independent 4-hour/30m rule tied to post views.
- Physical QR placement matters operationally: accuracy depends entirely on the geofence layer being correctly drawn per venue at onboarding.

17. Analytics

17.1 Validation metrics (network-level, tied to the MVP hypotheses in §4.1)

Metric	Definition
post_impressions	A post rendered within a user's viewport
post_push_notifications	Pushes sent referencing the post
post_opens	Full post detail screen opened
venue_profile_opens	Venue profile screen opened
navigation_clicks	"Navigate" tapped, attributed to source post/venue
qr_scans	QR resolutions across all three distance tiers
arrival_confirmations	Confirmed visits per §10.8
trial_started / trial_convert / promo_redeemed <i>(new)</i>	Subscription funnel events per §4.1 hypothesis 3

All are sourced from the append-only analytics/subscription event streams (§14.2–§14.3) — no metric is stored as an isolated counter.

17.2 Post Performance Card (venue-facing, per post)

Field	Computation
Views	count(post_impression) for the post
Push Opens	count(push_opened) where the push bundle included the post
Shares	count(share) for the post
Navigation Clicks	count(navigation_click) for the post
Confirmed Visits	count(arrival_confirmation) for the post
Conversion Rate	Confirmed Visits ÷ Views
Trend	sparkline of daily Views + Confirmed Visits over the post's active life

17.3 Venue dashboard (5 visuals max, per source brief's explicit cap)

1. **Headline metric** — **"Customers Brought In"**: count of arrival_confirmation events for the venue in the selected period, rendered above and larger than the five visuals below. (*Accessible only while the venue's subscription is Trial or Premium — REQ-SUB-007, §17.5.*)
2. **Post Views Over Time** — line chart.
3. **Push Performance funnel** — Sent → Open → Visit.

4. **Post Engagement** — bar chart of views/shares/navigation clicks per post.
5. **Visits from Posts** — conversion chart.
6. **Post Performance Card** — detail card for the selected/most recent post (§17.2), counted as the 5th visual.

Delegated team member view (*expanded, v1.5; Premium-gated, v1.6*): a venue’s invited team member/content creator (§5.3, §10.15) sees the same headline metric and four network-level charts as the owner, plus the Post Performance Card scoped to posts they personally authored — but not other creators’ individual post detail. All subscription/billing screens remain venue-owner-only (REQ-ACC-033–034, REQ-SUB-020). Like the rest of this dashboard, this view is only available while the venue is in Trial or on paid Premium; it is suspended (not removed) if the venue’s plan lapses to Free, and restores automatically on return to Premium (REQ-ACC-039–040).

17.4 Admin (platform) dashboard

Group	Metrics
Network health	venues posting in-network, posts per venue, views per post, venues with highest post conversion
Trending	venues ranked by cumulative engagement score (§10.6)
Push health	push open rates by area/neighborhood
QR health	QR scan volume by distance tier, scan time-of-day distribution
Growth	follower growth by venue
Platform/PWA	Core Web Vitals, PWA installs, notification/location opt-in rates, repeat visitor rate
Subscription health (<i>new</i>)	active trials, trial → paid conversion rate, MRR (monthly recurring revenue), Founding Venue slots used/remaining, promo-code redemptions, past-due count
Account/content operations (<i>new</i>)	onboarding-queue volume (new NYC venues, REQ-ADMIN-017), open report count, average time-to-resolution on reports

17.5 Refresh cadence & access

- Venue dashboard metrics shall refresh at least every 5 minutes.
- Admin dashboard metrics may refresh on a longer cycle (hourly) except the report-queue and onboarding-queue views, which shall reflect new entries immediately.
- The venue dashboard (§17.3) shall check the venue’s current subscription state (§21.2) on every load; a Free-plan venue shall be shown an upgrade prompt in place of the dashboard rather than a 403 with no explanation.
- Metabase (or an equivalent BI layer reading the analytics event stream) is the presentation layer for both dashboards.

18. Security

18.1 Authentication & session management

- JWT-based authentication (access + refresh token pair); access tokens shall be short-lived (≤ 15 min) with refresh rotation.
- Social sign-in (Google, Apple) shall be verified server-side against the provider's token, never trusted from client claims alone.
- "Remain signed in" sessions are capped at 30 days (REQ-ACC-007); refresh tokens shall be revocable server-side.
- Passwords shall be hashed with a modern adaptive algorithm (bcrypt or argon2), never reversibly stored.
- **New, v1.4:** the Admin portal (§7.8, §21's counterpart in account management) uses a **separate authentication path** (admin_users, §14.2) from the Firebase-backed consumer/venue auth described above, secured additionally by mandatory TOTP MFA (REQ-ADMIN-011). This is a deliberate isolation: a compromise of consumer auth infrastructure should not, by itself, grant any path into the Admin portal.

18.2 Role-based access control (RBAC)

Role	Scope
User	Own account, own follows, own device registrations
Venue Owner	Full venue account: profile, posts, staff invites, dashboard, subscription/billing (new)
Social Media Manager (Venue Team Member / Social Content Creator seat, one per venue, §10.15)	Create posts; edit/delete only posts they personally created, for the single venue they were invited to; view the venue's overall dashboard plus metrics scoped to their own posts (expanded, v1.5); no subscription/billing access (REQ-SUB-020, REQ-ACC-033); Trial/Premium-only — suspended (not deleted) if the venue is on the Free plan (REQ-ACC-039–040, new v1.6)
Admin	Report queue, content takedown, fraud queue, static pages, network dashboard, and — new, v1.4 — full view/edit/delete on Users, Venues, and Posts (REQ-ADMIN-014–016), authenticated exclusively through the dedicated MFA-protected portal

RBAC checks shall be enforced server-side on every request; the API layer shall reject out-of-scope requests with 403, not silently filter.

18.3 Bot & traffic protection

- Cloudflare Turnstile on all account-creation and login forms, **including the Admin portal login (REQ-ADMIN-012), which shall carry a stricter rate limit than standard user/venue login.**
- Cloudflare Bot Management and Rate Limiting in front of all public endpoints, with tighter limits on auth, claim, report, **checkout, and admin-portal endpoints specifically** (highest-abuse-value targets).
- Cloudflare provides HTTPS termination, DDoS mitigation, and WAF rules in front of both the Vercel-hosted frontend and the NestJS API.

18.4 Application security baseline

- Input validation and sanitization on every write endpoint; parameterized queries/ORM only.
- File upload validation via MIME-type and magic-byte checking.
- Secure token storage: refresh tokens in httpOnly, secure, SameSite=Strict cookies; access tokens held in memory client-side.
- Standard security headers (CSP, X-Content-Type-Options, Referrer-Policy, Strict-Transport-Security) on all responses.
- Dependency scanning shall run in CI and block merges on critical/high vulnerabilities.
- **New, v1.4:** Paddle webhook payloads (§12.6, §13.10) shall be signature-verified before any subscription state change is applied; unsigned or invalid-signature webhook requests shall be rejected and logged, never silently processed.

18.5 Push abuse prevention

- FCM token validation and per-device push caps (§15.4) double as abuse controls against notification-bombing.
- Push payloads shall never include PII beyond what's needed to render the notification.

18.6 Privacy & compliance

- Cookie consent (REQ-UX-040) meets GDPR (EU), CCPA (California), and general US disclosure norms.
- Location data is used transactionally and is not retained as a continuous trail.
- Drink-special compliance content (§10.2) and the age-gating flow (§10.1) are legal-compliance surfaces; final copy is ready to paste into static pages, pending Product's confirmation pass.
- Admin actions are fully audited (admin_audit_log, §14.2) to support accountability and incident review.
- **New, v1.4:** Nearby Vibes shall never store raw payment card data. Paddle.com acts as Merchant of Record and handles PCI-DSS scope, tax calculation/remittance, and card data entirely off-platform; the system stores only Paddle's subscription/transaction references (subscriptions.paddle_subscription_id) and derived plan-state data.

18.7 Reporting & incident handling

- The report pipeline (§10.9) doubles as an abuse-response mechanism; Admin takedown is available within the same session a report is filed, now with the inline view/edit/delete actions in REQ-ADMIN-019.
- A documented incident response runbook (credential compromise, data exposure, abuse spike, **and — new — payment/webhook anomalies**) is a pre-launch deliverable, tracked outside this PRD but referenced here as a gate.

19. Performance & Non-Functional Requirements

19.1 Scale targets

Parameter	Target
Concurrent supported users	10,000+
Supported venues	200+
Launch city	Manhattan, NY
Discovery radius	0.75 mi / 1.2 km default

19.2 Response time & availability budgets

Surface	Budget
API p95 response time (discovery, post read)	≤ 300 ms
API p95 response time (write endpoints: post publish, follow, claim, checkout initiation)	≤ 600 ms
Tile/Map initial paint (mobile, 4G)	≤ 2.5 s (Largest Contentful Paint)
Uptime target (core discovery + posting path)	99.5% monthly
Push delivery latency (bundling window aside)	≤ 60 s from FCM handoff to device receipt
Paddle webhook processing latency (<i>new</i>)	≤ 5 s from webhook receipt to subscription-state activation

19.3 Media performance (recap, authoritative detail in §7.5)

Rule	Value
Max pre-compression image upload	5 MB
Max pre-compression video upload	50 MB
Post-compression thumbnail	~20–50 KB
Post-compression feed image	~80–150 KB
Post-compression full image	~200–400 KB
Format	WebP primary; AVIF optional if needed later
Vercel image optimizer	Disabled entirely; custom Cloudflare-backed loader used instead

19.4 Caching & delivery architecture

- Media served exclusively from a dedicated media subdomain, never the app domain.
- Aggressive edge caching (Brotli, HTTP/3, ≥ 1 week CDN TTL on images) is mandatory, not optional.

19.5 Database performance

- PostgreSQL with PostGIS; all geofence/proximity queries shall use spatial indexes (GiST).
- The analytics event stream shall be indexed on (event_type, venue_id, occurred_at) and (event_type, post_id, occurred_at).
- **New, v1.4:** the subscriptions table shall be indexed on (state, trial_ends_at) and (state, current_period_end) to support the scheduled jobs that drive trial-expiry downgrades (§21.3) and renewal reminders (§15.9) at 10,000-venue scale without full-table scans.

19.6 Non-functional requirements

Requirement	Detail
Browser/PWA compatibility	Safari (iOS) and Chrome (Android/desktop) fully supported
No background location tracking	Location read only when foregrounded and a relevant check is triggered
Device-based tracking only	No cross-device identity resolution in MVP
Accessibility	WCAG 2.1 AA (§9.15)
SEO/indexing	Per §7.9; Admin portal path excluded (REQ-SEO-005)
Load testing	A load test simulating 10,000 concurrent users and 200 venues' worth of posting/discovery traffic shall be run against staging before launch, and shall include a simulated trial-expiry batch job run and a burst of concurrent Paddle checkout sessions.

20. Deployment

20.1 Final MVP stack

Layer	Choice	Rationale
Frontend	Next.js (React), hosted on Vercel	ISR + edge caching
Backend API	NestJS, hosted on DigitalOcean	Predictable compute cost at MVP scale
Database	PostgreSQL + PostGIS (DigitalOcean Managed Database)	Native geospatial queries
Object storage	Cloudflare R2 (or DigitalOcean Spaces as fallback)	Cheap egress
CDN / edge	Cloudflare	Caching, security, bot mitigation
Auth (consumer/venue)	Firebase Auth	Google/Apple/email flows
Auth (Admin portal)	Separate, MFA-protected auth service backed by admin_users (new)	Isolates operations access from consumer auth surface (§18.1)
Push	Firebase Cloud Messaging (Web Push)	Cross-platform push delivery
Maps (discovery + geofencing)	Mapbox GL JS	Geofence/polygon rendering, marker-icon support
Maps (venue claim import)	Google Places API	Required for the claim flow
Payments (new)	Paddle.com (Merchant of Record)	Handles global tax/compliance and PCI scope entirely off-platform (§18.6)
BI / dashboards	Metabase	Avoids building custom charting infra
Media processing	Sharp (Node), invoked once at upload time	One-time compute cost

20.2 Domain structure

Domain	Purpose
nearbyvibes.com	Main app (Vercel)
cdn.nearbyvibes.com / media.nearbyvibes.com	Media delivery only — never serves app HTML
api.nearbyvibes.com	NestJS backend, fronted by Cloudflare
[non-public path/subdomain — provisioned as a secret, not published here] (new)	Admin portal (REQ-ADMIN-010, §10.14). The literal value is intentionally withheld from this document; see §10.14 for why.

20.3 Environments & CI/CD

- Three environments: development, staging, production.
- CI pipeline shall run on every PR: type-check, lint, unit tests, dependency vulnerability scan, and a build check for both frontend and backend.
- Deploys to production shall be a manual promotion step gated on the load test and the acceptance criteria in §8 passing against staging, **including §8.13 (Subscription & Billing) and §8.14 (Admin Panel)**.
- Secrets (Firebase keys, Google Places API key, Cloudflare tokens, R2 credentials, database URLs, **Paddle API keys/webhook secrets, and the Admin portal path**) shall be stored in the hosting platforms' native secret managers, never committed to the repository.

20.4 Monitoring, logging, and alerting

- Application logs centralized to a log aggregator with retention of at least 30 days.
- Uptime monitoring against the 99.5% target on the core discovery, post-read, and post-publish endpoints.
- Push delivery failures and analytics event-ingestion failures shall be logged distinctly.
- **New, v1.4:** Paddle webhook failures (signature-verification failures, processing errors) and Admin-portal lock-out events (REQ-ADMIN-013) shall be logged distinctly and alerted to on-call, since both are high-signal for either a billing outage or an active credential-attack attempt.

20.5 Backup & disaster recovery

- Automated daily PostgreSQL backups with point-in-time recovery, retained a minimum of 14 days.
- Object storage is inherently durable via the provider; no separate media backup process required.
- A documented restore drill shall be run at least once before production launch.

20.6 Rollout plan

1. Internal/staging validation against all §8 acceptance criteria.
2. Load test at target scale (§19.6).
3. Manhattan-only launch.
4. Post-launch monitoring window (recommend minimum 2 weeks) against the three MVP validation hypotheses (§4.1) — **including trial-to-paid conversion** — before scoping the next iteration.

20.7 Forward-compatibility guardrails (non-blocking for MVP, binding on architecture choices)

- Auth (Firebase) and the core data model shall not assume “PWA” as a permanent constraint.
- The analytics event stream shall be platform-agnostic in schema.
- The `follows` and `posts` tables shall not encode any assumption that prevents future entitlement changes from being additive.
- **New, v1.4:** the `subscriptions` schema shall not assume exactly two paid tiers permanently — `plan_type` is an enum specifically so additional tiers (e.g., a future “Pro” tier) can be added as new enum values and `promo_codes` rows rather than a schema rewrite, consistent with this section’s original intent (§4.3 confirms multi-tier pricing itself remains out of scope for *this* MVP).

21. Subscription & Billing (*new chapter, v1.4*)

This chapter is the deep-dive companion to the catalog entries in §7.12, in the same structural pattern as §15 (Notification Engine) and §16 (QR Flow) relative to their own §7 catalog entries.

21.1 Purpose & validation link

Subscription & Billing exists to answer MVP validation hypothesis 3 (§4.1): whether the trial-to-paid mechanism converts venues into paying Premium customers, and whether the Founding Venue Program/promo mechanism meaningfully accelerates early Manhattan adoption without permanently devaluing the product (source brief, Promo Code section). It is deliberately the *smallest* billing system that can answer that question — see §4.3 for what remains out of scope (multi-tier pricing, seat-based pricing, full dunning/win-back automation, in-app self-service cancellation beyond a basic cancel action).

21.2 Subscription state model

State	Meaning	Entered from	Exits to
trial	14-day, no-card-required Premium access	Venue claim completion	active (checkout completes) or free (14 days elapse)
free	Default plan; Premium features suppressed	Trial expiry or subscription non-renewal	active (checkout completes at any time)
active	Paid Premium (Monthly or Annual), full feature access	Trial upgrade, or checkout from Free	past_due (renewal payment fails) or cancelled
past_due	Payment failed; grace window in effect	active, on failed renewal webhook	active (retry succeeds) or free (grace window elapses)
cancelled	Venue-initiated cancellation, effective at period end (<i>P1</i>)	active	free, once current_period_end passes

PM Decision: the source brief did not specify a grace-window duration for a failed renewal payment. This document sets a **default 3-day grace window** between a past_due state and automatic downgrade to free, giving Paddle’s own retry logic and the venue owner time to update payment details, consistent with common SaaS practice. This value should be treated as configuration, not a hardcoded constant, and confirmed by Product before launch.

21.3 Scheduled jobs

- **Trial-expiry sweep:** runs at least hourly; selects subscriptions rows where `state = trial` and `trial_ends_at <= now()`, transitions them to `free`, and fires the trial-expired notification (§15.9).
- **Renewal-reminder sweep:** runs daily; selects active subscriptions with `current_period_end` within 3 days, fires the renewal-upcoming notification (§15.9).
- **Past-due grace sweep:** runs at least hourly; selects `past_due` subscriptions past their grace-window deadline, transitions them to `free`, and fires the subscription-expired notification (§15.9).
- **Founding Venue cap check:** evaluated synchronously at checkout time (not as a batch job) against `promo_codes.redemption_count < max_redemptions`, since this must be race-condition-safe under concurrent checkouts (see §21.4).

21.4 Checkout, Paddle integration, and concurrency

Checkout is initiated per §12.6 and REQ-SUB-011–REQ-SUB-015. Two implementation notes that materially affect correctness:

- **Founding Venue cap concurrency:** because multiple venues can attempt to redeem the last remaining Founding Venue slot simultaneously, the redemption-count increment (§14.2, `promo_codes.redemption_count`) shall be performed as an atomic, row-locked database operation at the point of successful payment confirmation (the Paddle webhook, not at checkout-session creation), so that a venue who starts checkout against an available slot but pays after the cap fills is not incorrectly granted the Founding Venue benefits. If this occurs, the system shall still complete the underlying paid subscription at the standard promo discount (if the general promo code is still valid) or standard price, and shall notify the venue that Founding Venue status specifically was unavailable — never fail the payment itself over a benefits-eligibility race.
- **Paddle as Merchor of Record:** Paddle handles the actual payment method storage, tax calculation, and remittance. Nearby Vibes' backend only ever sees a `paddle_subscription_id` and webhook-delivered state changes — never raw card data (§18.6).

21.5 Auditability

Every subscription state transition — trial start, trial expiry, upgrade, downgrade, promo applied, renewal, payment failure, cancellation — shall be written to `subscription_events` (§14.2) before or atomically with the corresponding subscriptions row update, so that a venue's billing history is always reconstructable independent of the current-state row (REQ-SUB-018), mirroring the same append-only philosophy already used for the analytics event stream (§14.3) and the admin audit log (§14.2).

21.6 Relationship to Admin

Admins can view (but the MVP does not require them to directly edit) a venue's subscription state from the Venue account-management screen (REQ-ADMIN-015, §9.12) — e.g., to confirm a support-reported billing issue — with any correction made going through the standard checkout/webhook path rather than a raw database edit, to preserve the audit trail in §21.5.

Appendix A — Decisions Log

Version 1.0 of this document raised five open items requiring Product sign-off before build; all five were confirmed by v1.3 (items 1–5 below, unchanged). The 23 July Additional Requirements brief raised two further scope-level items, resolved in v1.4 as items 6–7. Same-day follow-up notes raised three further items, resolved in v1.5–v1.8 as items 8–10. As of v1.8, no open items remain.

#	Item	Status	Resolution
1	QR URL scheme (§16.1)	Confirmed	Single universal QR code (nearbyvibes.com/qr), no per-venue identifier. Venue resolved server-side via geofence into three distance tiers.
2	Engagement scoring (§10.6)	Confirmed	Point values as listed in §10.6; 2-hour batch cycle; 5-point conversion threshold within the 48-hour trial window.
3	Legal copy (§10.2, §18.6)	Confirmed — content ready	Terms, Privacy, AUP, and disclaimer copy exist and are ready to paste into static pages; content-population task, not a drafting task.
4	Map provider (§20.1)	Confirmed	Mapbox GL JS for rendering/visualization; PostGIS remains server-side source of truth for geofence containment.
5	Visual design pass (§9.1, §9.16)	Confirmed — direction supplied, execution in progress	Direction locked: on.com/zillow.com/alltrails.com/nps.gov references, confirmed brand palette. Executed via the companion Design Brief.

#	Item	Status	Resolution
6	Subscription & Billing scope (§4.2–§4.3, §7.12, §21)	Confirmed, v1.4	The 23 July Additional Requirements brief moves subscriptions and Paddle billing from “explicitly out of scope, deferred to a second iteration” (v1.0–v1.3) into the MVP itself: a 14-day no-card trial, Free/Premium (\$39/mo or \$390/yr) plans, a Founding Venue Program (first 50 Manhattan venues) and general promo-code mechanism, Paddle-processed checkout with a disclosed 5%+\$0.50 platform service fee, and pre-expiry/renewal lifecycle notifications. Two sub-items were ambiguous in the source brief and are flagged as open PM Decisions rather than silently resolved: (a) whether the Founding Venue Program and the general “qualifying NYC venues” Promo Code are the same offer or two related-but-distinct offers (§10.12 treats them as distinct, sharing a data model); and (b) the exact trial-start trigger instant, set here to venue-claim completion rather than profile completion (§10.11). Product should confirm both before build sign-off.

#	Item	Status	Resolution
7	Admin Panel security & scope (§7.8, §18)	Confirmed, v1.4	The Admin Panel gains a dedicated, MFA-protected authentication portal at a non-public URL (path itself withheld from this document, §10.14, §20.2); full view/edit/delete control over Users, Venues, and Posts; an alert when a new, fully onboarded venue resolves within New York City; and a linked, in-place view/edit/delete workflow on every content report. This supersedes the v1.0–v1.3 Admin scope, which covered dashboard visibility and basic report takedown but not full account management or a separated authentication surface.

#	Item	Status	Resolution
8	Delegated Venue Seat — role types, seat cap, dashboard scope (§5.3, §7.1, §10.15)	Confirmed, v1.6	A venue owner may invite exactly one team member/content creator per venue, by email, choosing a role type (Venue Team Member or Social Content Creator) and a custom title before the invite is sent. The invited person creates their own account; may create posts and edit/delete only posts they personally created; and can view the venue's overall performance dashboard in addition to their own post detail — an expansion from the prior “own posts only” dashboard restriction (REQ-ACC-034). Update, v1.6: Product has confirmed that Venue Team Member and Social Content Creator share one identical privilege set, with the role type serving purely as an organizational label (§10.15) — this is no longer an open item.

#	Item	Status	Resolution
9	Delegated Venue Seat gated to Trial/Premium subscription (§7.1 REQ-ACC-039–040, §10.15)	Confirmed, v1.7	Raised by Product alongside the item 8 confirmation: the delegated seat is a Premium-tier capability, available during the 14-day Trial or on an active paid Premium subscription, but not on the Free plan. A venue on Free cannot send a new seat invite. If an existing seat's venue downgrades to Free, the seat's access is suspended (not deleted) — the seat-holder loses post/dashboard access while the venue is on Free, and access is restored automatically, with no re-invite needed, if the venue returns to Trial or Premium. Product has confirmed suspend-and-restore is the intended behavior (not a harder revoke-and-re-invite); this item is now fully closed.

#	Item	Status	Resolution
10	Trial & re-claim abuse prevention via Google Place ID (§7.2 REQ-ONB-023–026, §7.12 REQ-SUB-021, §10.16)	Confirmed, v1.8	Closes a gaming pattern: exhausting the 14-day trial, deleting the venue account, and re-claiming the same physical venue under a new email to draw a second trial. Two linked, confirmed constraints: (1) a Google Place ID is entitled to exactly one 14-day free trial, ever and permanently — tracked independent of account deletion via a persistent <code>place_claim_ledger</code> (§14.2), and never resets; and (2) separately, a deleted Place ID cannot be re-claimed by any account for a configurable cooldown, default 90 days (~3 months), with an Admin override for legitimate ownership transfers. Update, v1.8: Product has confirmed both the permanence of the trial restriction and the claim-only scope of the 90-day cooldown exactly as specified — this item is fully closed with no remaining open questions.